

PRIMARY 6 BASIC DIGITAL LITERACY SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	Advanced Computer Systems	Computer architecture overview. Binary system basics: how computers use 0s and 1s. Bits and bytes: measuring data. Processing cycle: input, processing, output, storage. Computer performance factors: CPU speed, RAM, storage type.
2	Computer Networks	Understanding networks: connected computers sharing resources. Types: LAN, WAN, Internet. Network components: router, modem, switch, cables, WiFi. IP addresses basics. Benefits of networking: resource sharing, communication, collaboration.
3	Internet and Web	Internet vs World Wide Web difference. How internet works: packets, servers, browsers. URL structure: protocol, domain, path. Web browsers: Chrome, Firefox, Edge, Safari. Search engines: effective searching techniques. Web vs mobile apps.
4	Cloud Computing	Understanding cloud computing: using internet-based services. Cloud storage: Google Drive, Dropbox, OneDrive, iCloud. Benefits: access anywhere, backup, collaboration. Cloud applications: Google Docs, Office 365. Security considerations. Managing cloud storage.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Advanced Word Processing	Professional document creation. Advanced formatting: styles, themes, columns. Mail merge basics. Comments and track changes. Inserting citations and footnotes.

		Table of contents automation. Creating forms.
8	Advanced Presentations	Professional presentation skills. Master slides and templates. Multimedia integration: video, audio. Advanced animations and transitions. Hyperlinks within presentations. Presenter notes and timing. Recording presentations.
9	Advanced Spreadsheets	Complex formulas: IF statements, COUNT, MAX, MIN. Cell referencing: absolute vs relative. Creating charts and graphs. Data sorting and filtering. Conditional formatting. Using spreadsheets for budgets and planning.
10	Database Basics	Introduction to databases: organized information collection. Database structure: tables, records, fields. Creating simple databases. Sorting and filtering data. Queries basics. Databases vs spreadsheets. Practical applications.
11	REVISION	Review of all topics. Complete integrated projects using multiple applications.
12	EXAMINATION	End of First Term Examination (Theory and Practical)
13	CLOSING	Report distribution and vacation

PRIMARY 6 DIGITAL LITERACY LESSON NOTES

FIRST TERM

WEEK 1: ADVANCED COMPUTER SYSTEMS

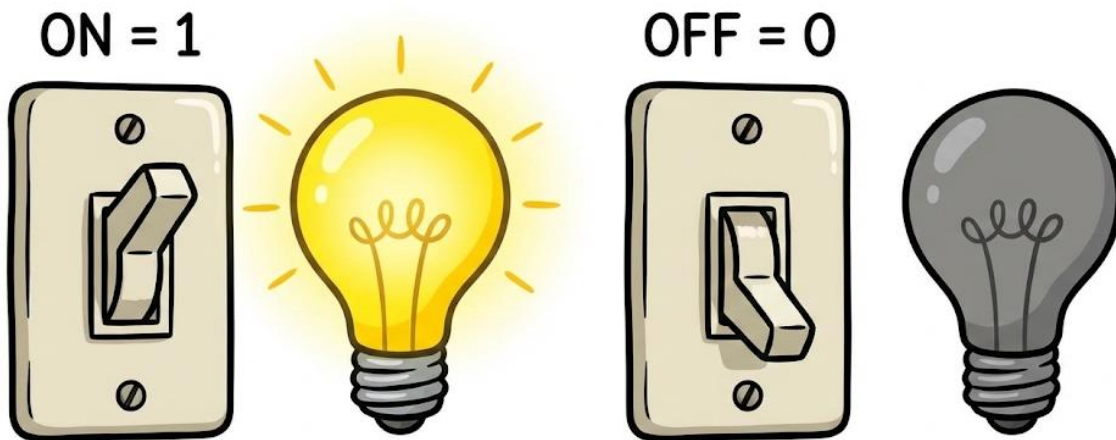
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the basic concept of the binary system (how computers count).
2. Differentiate between bits and bytes as units of data measurement.
3. Describe the four stages of the computer processing cycle.
4. Identify factors that affect the speed and performance of a computer.

CONTENT

1. The Binary System (Computer Language)

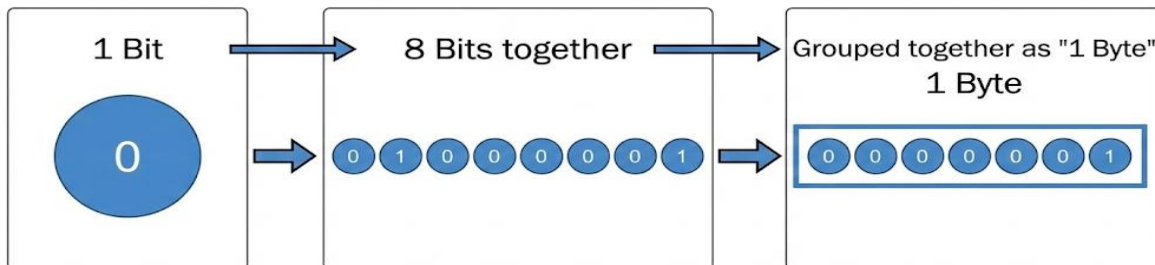
- Humans count using ten symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. This is called the *decimal system*. We use it for everything, like counting pencils, money, or our age.
- Computers are electronic machines. They don't have ten fingers. They only understand electricity. Electricity can be in two simple states: **ON** (power is flowing) or **OFF** (power is not flowing).
- To talk to computers, we use their language. We use **0** to mean **OFF** and **1** to mean **ON**. This language with only two numbers (0 and 1) is called the **Binary System**.
- Every single thing a computer does—showing a letter, playing a song, drawing a picture—is made from very long combinations of just 0s and 1s. It's like a secret code.
 - **Example:** The letter 'A' in computer language (binary) is 01000001.



Computers understand only ON (1) and OFF (0),
like a light switch.

2. Bits and Bytes

- **Bit:** This is the smallest piece of computer data. The name comes from **B**inary **D**igit. One bit is a single **0** OR a single **1**.
 - It's like a single bead in a necklace.
- **Byte:** A group of **8 bits** together makes **1 Byte**. One byte is powerful enough to store one character, like the letter '**B**', the number '**5**', or a symbol like '**\$**'.
 - It's like stringing 8 beads together to make one complete pattern.



- **Measuring Data:** We measure digital information just like we measure weight in kilograms or distance in meters.
 - **1 Byte** = One character (e.g., 'A')
 - **1 Kilobyte (KB)** = About 1,000 bytes. (Like a short paragraph of text).
 - **1 Megabyte (MB)** = About 1,000 Kilobytes. (Like a 3-minute song from your phone).
 - **1 Gigabyte (GB)** = About 1,000 Megabytes. (Like a 30-minute video or a computer game).

A SIMPLE TABLE COMPARING DATA SIZES WITH FAMILIAR NIGERIAN EXAMPLES

Unit	How Many Bytes?	What It Can Hold (Example)
Bit	Tiny piece	A 0 or a 1
Byte (B)	8 bits	One letter (e.g., 'C')
Kilobyte (KB)	~1,000 bytes	A simple, small picture
Megabyte (MB)	~1,000 KB	One popular Nigerian song
Gigabyte (GB)	~1,000 MB	A Nollywood movie on a phone

3. The Processing Cycle (IPOS)

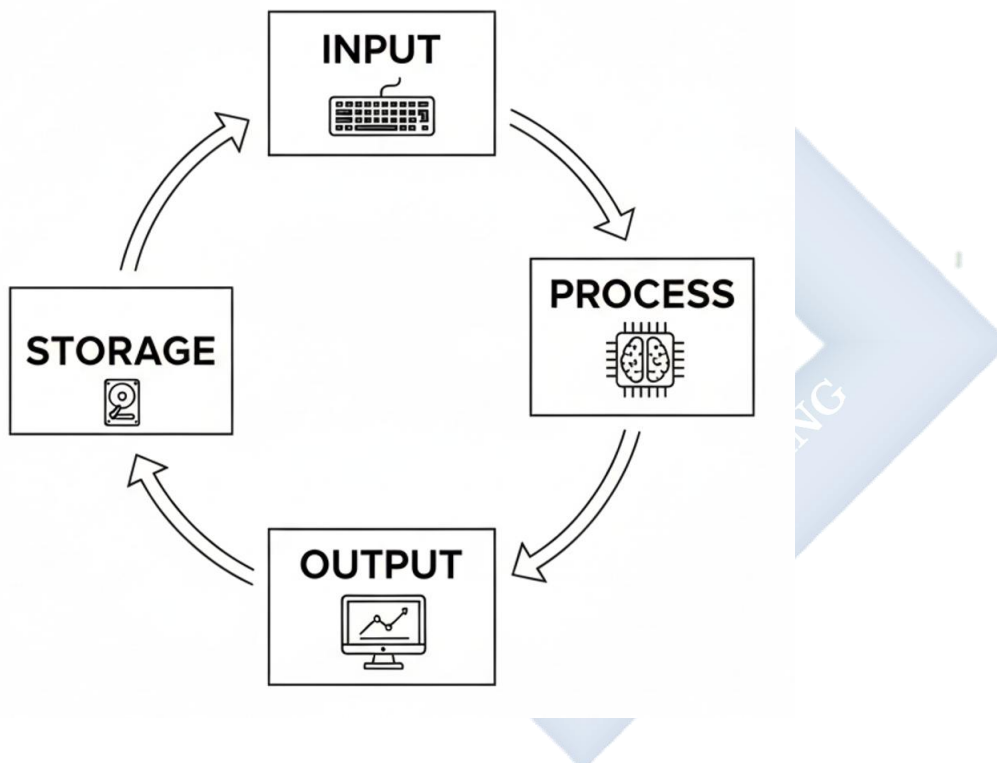
- A computer works in a never-ending four-step circle called the **IPOS Cycle**.
 - INPUT:** This is how we give the computer information or instructions. We use **input devices**.
 - *Examples:* Typing with a **keyboard**, pointing and clicking with a **mouse**, speaking into a **microphone**, or taking a picture with a **webcam**.
 - PROCESSING:** This is the computer's thinking stage. The **CPU (Central Processing Unit)**, the computer's brain, takes the input and works on it. It calculates, organizes, and makes decisions.
 - *Example:* When you type $5 + 3$ and press '=', the CPU calculates the answer is 8.

- c. **OUTPUT:** This is how the computer shows or gives us the result of its processing.
We use **output devices**.

- *Examples:* Seeing the answer '8' on the **monitor**, printing your work on a **printer**, or hearing sound from **speakers**.

- d. **STORAGE:** This is how the computer saves the work for later. It keeps information even when the power is off.

- *Examples:* Saving your work on the computer's **hard drive**, on a **USB flash drive**, or in **cloud storage** online.



4. Computer Performance Factors

- Why are some computers faster than others? It depends on their parts, just like a car's speed depends on its engine.
 - **CPU Speed (The Brain):** Measured in **GHz (Gigahertz)**. A higher number (e.g., 2.5 GHz) means the CPU can "think" and calculate faster.
 - **RAM (Random Access Memory):** This is the computer's **short-term memory** or "work desk." More RAM (e.g., 8GB instead of 4GB) means the computer can

have many programs (like a browser, Word, and a game) open at once without becoming slow or freezing.

- *Analogy:* A bigger desk (more RAM) lets you spread out more books and work on them easily without shuffling papers.

- **Storage Type (The Filing Cabinet):**

- **HDD (Hard Disk Drive):** Older, slower, uses spinning disks. Like finding a file in a big, crowded cabinet.
- **SSD (Solid State Drive):** Newer, much faster, has no moving parts. Like a magical cabinet that gives you any file instantly. Computers with SSDs start up very quickly.

EVALUATION

1. Computers use a language of two numbers. What are they?
2. If one Bit is a single '1', how many Bits are needed to make one Byte?
3. In the IPOS cycle, what happens during the 'Processing' stage?
4. Which part of the computer is called its "brain" and does all the processing?
5. Explain in your own words: Why does a computer with more RAM work better when you have many programs open?
6. Your father wants to buy a new computer that starts up very fast. Should you advise him to look for one with an HDD or an SSD? Why?

ASSIGNMENT

1. **(Oral/Practical):** Look around your home. Find and tell your classmate about **three** appliances that have simple ON/OFF switches (like a torch, fan, or radio). Explain how this is similar to a computer's binary system.
2. **(Written):** Arrange these units of computer data from the **smallest to the biggest**: Gigabyte (GB), Bit, Kilobyte (KB), Byte, Megabyte (MB).
3. **(Drawing/Written):** Think about making a cup of Milo. **Draw a simple 4-step picture** showing the IPOS cycle for this task. (E.g., Input = getting milk and Milo; Process =

mixing and stirring; Output = the finished drink; Storage = putting the Milo tin back in the cupboard).



WEEK 2: COMPUTER NETWORKS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a computer network.
2. Differentiate between LAN and WAN.
3. Identify common network hardware components.
4. State the benefits of networking computers together.

CONTENT

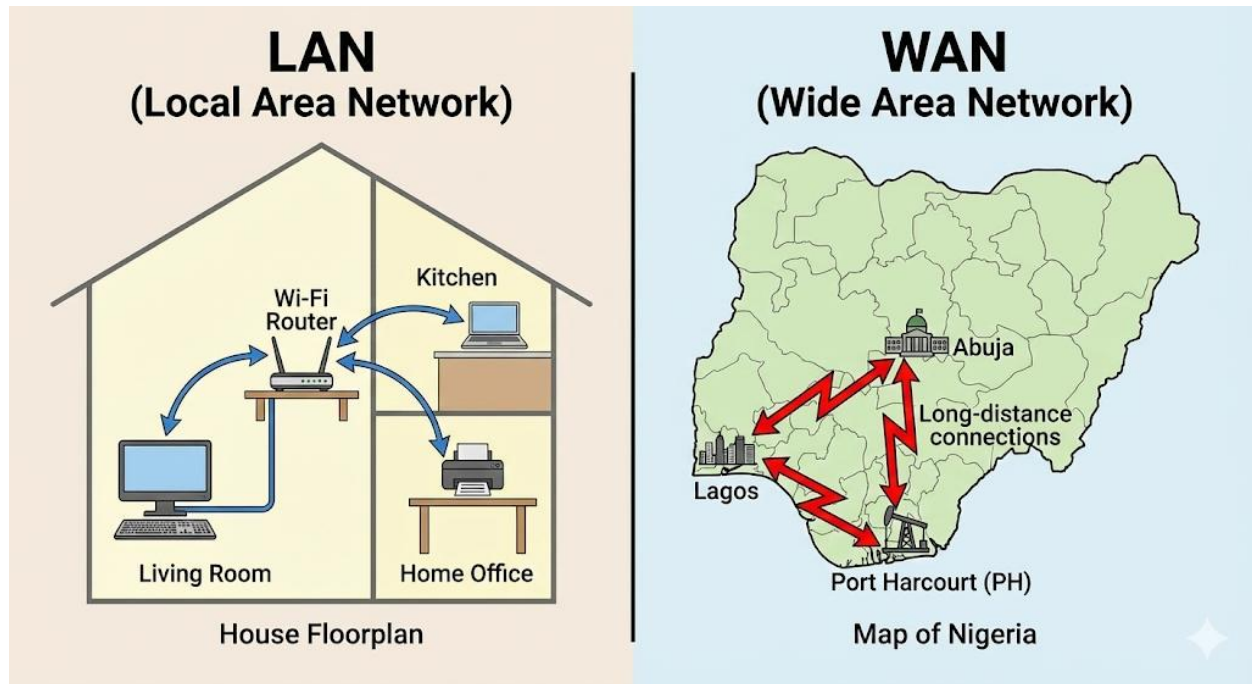
1. Understanding Computer Networks

- A **Computer Network** is created when two or more computers are connected together to share things. They can be connected with physical **cables** or through invisible **wireless signals** (like Wi-Fi).
- **Simple Idea:** It's like connecting houses in a village with roads. The roads allow people to visit and share things. In a network, computers can "visit" and share information with each other.
- **Main Purpose:** To share resources and communicate.

2. Types of Networks

- **LAN (Local Area Network):**
 - **Meaning:** Local Area Network.
 - **Covers:** A small, close area.
 - **Examples:** The computers in your **school's computer lab**, all the devices (computer, phone, smart TV) connected to the Wi-Fi in **your home**, or the computers in a **single office building**.
- **WAN (Wide Area Network):**
 - **Meaning:** Wide Area Network.
 - **Covers:** A very large geographical area.

- **Examples:** The **Internet** is the biggest WAN in the world. It connects countries and continents. A bank connecting its branches in Lagos, Abuja, and Port Harcourt also uses a WAN.



3. Network Components (Hardware)

- To build a network, we need special devices:
 - **Modem:** This device brings the internet connection **into** your building from your service provider (like Spectranet, MTN, or Airtel). It's the **gatekeeper** to the outside world (the Internet).
 - **Router:** This device takes the internet from the modem and **shares it** with all the devices in your home or school. It acts like a **traffic policeman**, directing data to the correct computer, phone, or printer. Most routers also create a Wi-Fi signal.
 - **Switch:** Used mostly in offices or labs. It has many ports to connect many computers together using **Ethernet cables** (those thick cables with a plug that looks like a big phone jack).
 - **Network Interface Card (NIC):** This is a part inside **every** computer (like a built-in part or a small added card) that allows it to connect to a network, either with a cable or wirelessly.

- **IP Address:** Every device on a network gets a unique number called an **IP Address**. It's like your home address for mail. When data is sent across the network, the IP address ensures it gets to the right computer. (e.g., 192.168.1.5).

4. Benefits of Networking

- **Sharing Hardware:** Many computers can use one **printer**, one **scanner**, or one internet connection. Saves money!
- **Sharing Files and Software:** You can send a project file to your friend's computer instantly without using a flash drive. Some software can be installed on one computer and used by others on the network.
- **Communication:** You can send messages, emails, or even have video calls with people on the same network.
- **Working Together (Collaboration):** Many people can work on the same document or project at the same time from different computers.

EVALUATION

1. What is a computer network?
2. Which type of network connects computers within one school building: LAN or WAN?
3. What is the main job of a **router** in a home network?
4. Why does every computer on a network need an IP address?
5. If your school has only one printer but 20 computers, how does a network help?
6. Name one device that brings the internet signal into a building from the service provider.

ASSIGNMENT

1. **(Drawing):** Draw a simple diagram of a **home network**. Show one modem connected to one router, and the router connected to three devices (e.g., a computer, a laptop, and a phone). Label each device.
2. **(Written):** Look at your school's computer lab. Is it an example of a LAN or a WAN? Write down your answer and give one reason why.

3. **(Oral/Written):** List **two** things (besides a printer) that students in a networked computer lab could share with each other.
-



WEEK 3: INTERNET AND WEB

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Distinguish between the Internet and the World Wide Web.
2. Identify the parts of a standard URL (web address).
3. List examples of popular web browsers.
4. Demonstrate effective techniques for searching information online.

CONTENT

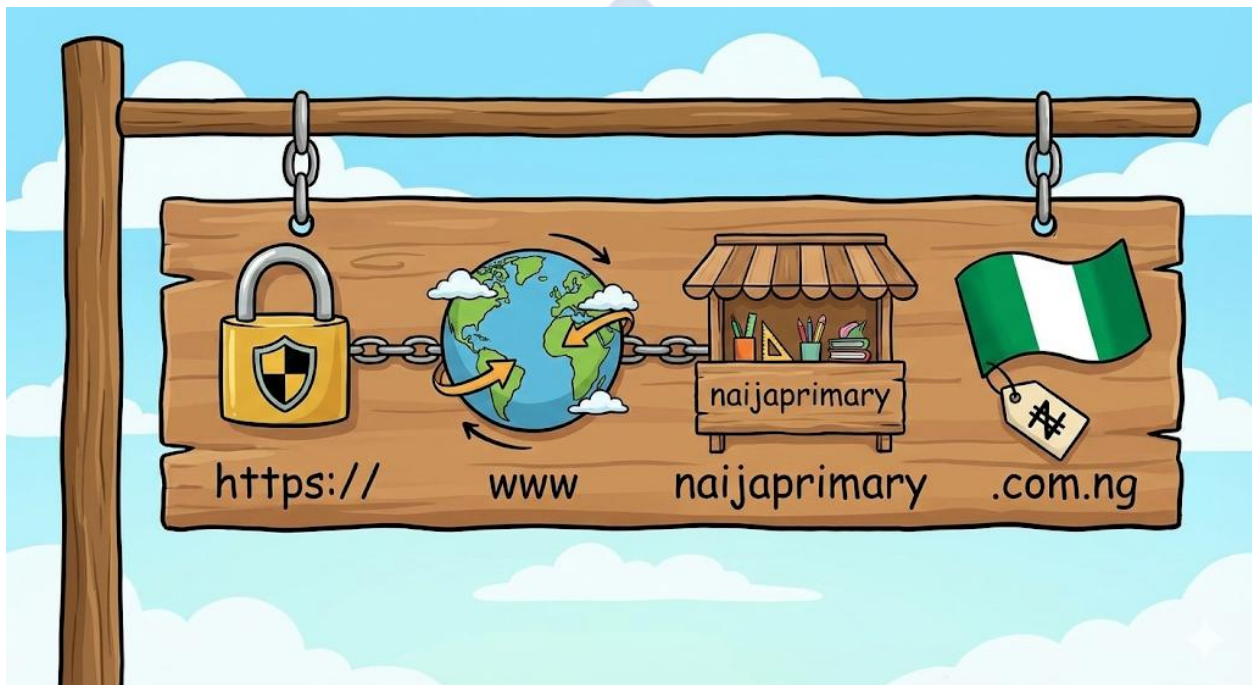
1. Internet vs. World Wide Web (They are NOT the same!)

- **The Internet:** This is the **giant physical connection** of millions of computers, cables (under the sea and on land), satellites, and towers all around the world. Think of it as the **global network of roads and highways**. It is the **infrastructure** that allows computers to talk.
- **The World Wide Web (WWW or "the Web"):** This is the **collection of all the websites, pages, pictures, and videos** that are stored on computers connected to the Internet. Think of it as all the **shops, markets, libraries, and cinemas** you can visit on those roads.
- **Simple Analogy:** The **Internet is the road**. The **Web is the bus** that carries information (websites) on that road. You need the road (Internet) for the bus (Web) to move.

2. Understanding a URL (Web Address)

- **URL** stands for **Uniform Resource Locator**. It is the full address of a specific webpage.
- Let's break down a common URL: <https://www.naijapprimary.com.ng>
 - **https://** – The **Protocol**. It's the set of rules for how the browser and the website talk. The 's' means it's a **secure** connection (good for logging in).
 - **www** – **World Wide Web**. This part tells you it's on the web.
 - **naijapprimary** – The **Domain Name**. This is the chosen name of the website (like a shop's name). It's easy for humans to remember.

- **.com.ng** – The **Domain Extension**. This tells you about the website's type or location.
 - .com = Commercial (business)
 - .org = Organization (non-profit)
 - .edu.ng = Educational (Nigerian school/university)
 - .gov.ng = Government (Nigeria)
 - .ng = Nigeria



3. Web Browsers

- A **Web Browser** is a **software application** you open on your computer or phone to access and view websites. **Without a browser, you cannot see the web.**
- **Common Examples:**
 - **Google Chrome** (Very popular, green-red-yellow icon)
 - **Microsoft Edge** (Comes with Windows computers, blue 'e' icon)
 - **Mozilla Firefox** (Orange fox wrapped around a globe)
 - **Safari** (Comes with Apple devices like Mac and iPhone, blue compass icon)

4. Search Engines and Smart Searching

- A **Search Engine** is a special website that helps you **find** other websites when you don't know the exact URL. **Google** is the most popular search engine. Others are **Bing** and **Yahoo**.
- **How to Search Effectively:**
 - a. **Use Specific Keywords:** Instead of searching "food," search for "**how to cook jollof rice Nigeria**." The more specific, the better the results.
 - b. **Use Quotation Marks for Exact Phrases:** If you want a specific quote or title, put it in quotes. E.g., "**No condition is permanent**" will find pages with that exact sentence.
 - c. **Check the Source:** Always look at the website's name. Information from .gov.ng, .edu.ng, or well-known news sites (like BBC) is usually more reliable than a random blog.

EVALUATION

1. True or False: The Internet and the World Wide Web mean exactly the same thing.
2. What does **URL** stand for?
3. Look at this address: <https://www.unilag.edu.ng>. What does the **.edu.ng** part tell you about the website?
4. Name two different web browsers.
5. If you wanted to find a story about "The Tortoise and the Hare," what would be a better search: (a) just "tortoise" or (b) "**Tortoise and the Hare story**"? Why?
6. What symbol do you put around words to search for an exact phrase?

ASSIGNMENT

1. **(Written):** Write down the **full web address (URL)** of **two** websites you or your parents use. Next to each, write what you think the domain extension means (e.g., .com for business).
2. **(Practical/Oral):** On a computer or phone (with permission), open a web browser. Tell your teacher or friend: **What is the name of the browser?** (Look at its icon).

3. **(Written):** You need to find information for a project on "Traditional Nigerian Festivals." Write down **three specific keyword phrases** you would type into a search engine.
-



WEEK 4: CLOUD COMPUTING

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define cloud computing in simple terms.
2. List examples of cloud storage services.
3. State the benefits of using cloud services.
4. Mention basic security measures for cloud accounts.

CONTENT

1. Understanding Cloud Computing

- **"The Cloud"** is a nickname for storing your files and using software **over the Internet** instead of on your own computer's hard drive.
- **Simple Analogy:** Imagine your computer's hard drive is the **bag you carry to school**. The Cloud is like a **big, safe locker at school**. You can put your books (files) in the locker and access them anytime you are in school (have internet). You can also share books from your locker with a friend.
- **Key Idea:** Your files are not *on your device*; they are on powerful computers called **servers** in special buildings (data centres) far away, and you access them via the Internet.



2. Cloud Storage Examples (Your Digital Lockers)

- These are free or paid services that give you space in the cloud:
 - **Google Drive:** Comes free with a Gmail account. Great for schoolwork.
 - **Microsoft OneDrive:** Comes with a Microsoft/Outlook account. Works well with Microsoft Word and PowerPoint.
 - **Apple iCloud:** Used mostly by people with iPhones and iPads to backup photos and messages.
 - **Dropbox:** A simple and popular service for storing any type of file.

3. Benefits of Cloud Computing

- **Access Anywhere:** You can open your project on your home computer, your school computer, or even your parent's phone, as long as you log into your account.
- **Safe Backup:** If your computer crashes, gets stolen, or your flash drive is lost, your files are still safe in the cloud.
- **Easy Sharing and Collaboration:** You can share a file with your group members by sending them a link. You can all work on the **same document** at the same time (like in **Google Docs**) and see each other's changes live.
- **Saves Device Space:** Large files like videos can live in the cloud, freeing up space on your phone or computer.

4. Cloud Applications and Security

- **Cloud Apps:** These are software you use directly in your web browser without installing anything. Examples: **Google Docs** (for typing), **Google Sheets** (for calculations), **Canva** (for designing posters).
- **Security is Very Important:** Since your files are online, you must protect your account.
 - a. **Strong Password:** Use a mix of uppercase letters, lowercase letters, numbers, and symbols. **Never** use easy passwords like "password123" or your name.
 - *Example of a strong password:* Lagos4Ever! (but don't use this exact one!).

- b. **Log Out on Shared Computers:** Always click "Log Out" when you finish using a cloud service on a school or café computer.
- c. **Beware of Sharing:** Be careful when sharing file links. Only share with people you trust.

EVALUATION

1. When we save a file to "the cloud," where is it actually saved?
2. Name two popular cloud storage services.
3. If you spill water on your laptop and it stops working, what is one big advantage of having saved your project in the cloud?
4. What is one cloud application you can use to type a letter?
5. Why is it dangerous to use a simple password like "flower" for your cloud account?
6. How does cloud computing help a group of students working on one project?

ASSIGNMENT

1. **(Oral/Research):** Ask an adult at home or a teacher: "Do you use any cloud service like Google Drive or iCloud?" Write down the name of the service they mention and **one thing** they use it for.
 2. **(Written):** List **three different types** of files that are good to save in the cloud (e.g., holiday photo album, Primary 6 project).
 3. **(Practical/Written):** Create a **strong, pretend password** for a new cloud account. Write it down. Explain to a friend why it is strong (mention if it has uppercase, numbers, etc.).
-

WEEK 5 & 6: MIDTERM EXAMINATION AND BREAK



WEEK 7: ADVANCED WORD PROCESSING

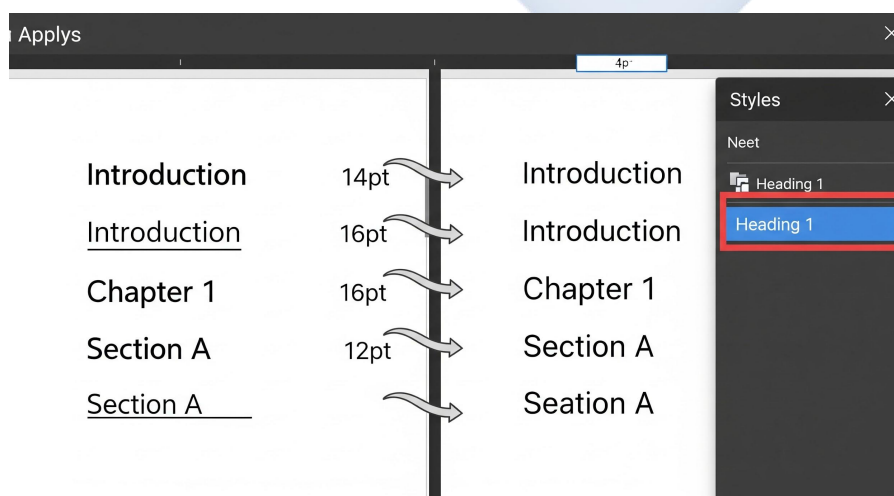
OBJECTIVES By the **end** of the lesson, pupils should be able to:

1. Apply advanced formatting such as styles and columns to a document.
2. Explain the purpose of the Mail Merge tool.
3. Use the Track Changes and Comments feature.
4. Insert a Table of Contents automatically.

CONTENT

1. Advanced Formatting

- **Styles:** This is a set of formatting instructions (font, size, colour) saved as one command. Instead of manually making every main heading bold, size 16, and blue, you just click the "**Heading 1**" style. It makes your document look neat, professional, and consistent.
 - *Try it:* Type "My School." Select it and apply the "**Title**" style. See how it changes.
- **Columns:** This splits your page text into vertical sections, like in a newspaper. Useful for leaflets or newsletters.
 - *Go to:* **Layout > Columns**. Choose Two or Three.
- **Themes:** This changes the overall look of your entire document—the colours of headings, the style of lines, and the font family—all with one click.



2. Mail Merge Basics

- **Purpose:** To create many personalized letters, envelopes, or labels from one template and a list of names/addresses.
- **Real-Life Example:** Your Head Teacher wants to send a "Congratulations on your promotion" letter to 50 parents. Instead of typing 50 separate letters, she types **one** main letter. She has a list of parent names in a table. Mail Merge **merges** the list with the letter, printing 50 copies, each with the correct parent's name inserted automatically.
- **Main Parts:** 1) The Main Document (the letter), 2) The Data Source (the list of names), 3) The Merged Document (the final letters).

3. Comments and Track Changes

- These are tools for **working with others** on one document.
- **Comments:** Allows you to stick a **digital note** on the side of the document to ask a question or make a suggestion **without changing the original text**. It's like putting a sticky note on a book page.
 - *To use:* Highlight text, go to **Review > New Comment**.
- **Track Changes:** When this is **ON**, every edit you make (deleting, adding, formatting) is recorded and shown in colour. Deleted text has a strikethrough, new text is underlined. Others can see exactly what you changed and choose to accept or reject your changes. It's like a teacher marking your assignment with a red pen.

4. Table of Contents

- A **Table of Contents (TOC)** is a list at the beginning of a long document (like a project or storybook) that shows the chapters/sections and their page numbers.
- **The Magic Trick:** If you have used **Styles** (like Heading 1, Heading 2) for your section titles, the word processor can **create the TOC automatically** and even update the page numbers if you add more pages.
 - *Go to:* **References > Table of Contents**. Choose a style.

EVALUATION

1. What is the main advantage of using "Styles" instead of formatting each heading by hand?
2. Where would you find the tool to split your text into two columns like a newspaper?
3. Imagine you are the class secretary and need to invite 30 parents to a meeting. Which word processing tool would save you from typing 30 separate invitation letters?
4. How is leaving a "Comment" different from just typing your suggestion into the document?
5. What is the purpose of a Table of Contents?
6. What must you do to your headings first, before the computer can create an automatic Table of Contents?

ASSIGNMENT

1. **(Practical/Observation):** Look at a newspaper, magazine, or a church bulletin. Find a part that uses **columns**. Write down the title of that article or section.
 2. **(Written/Planning):** You are writing a 3-page project about "My Local Government." Write down **three section headings** you would use and which **Style** (Title, Heading 1, Heading 2) you would apply to each.
 3. **(Written Explanation):** In one sentence, explain why "Track Changes" is very useful when a teacher is checking your digital assignment.
-

WEEK 8: ADVANCED PRESENTATIONS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the function of a Master Slide.
2. Insert multimedia (audio and video) into a presentation.
3. Differentiate between animations and transitions.
4. Create hyperlinks within a presentation.

CONTENT

1. Master Slides and Templates

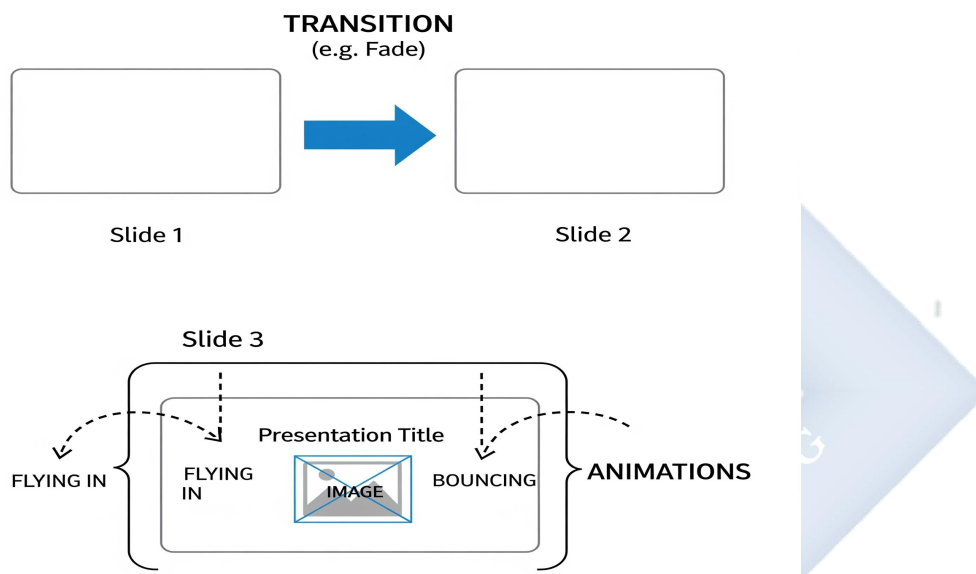
- **Master Slide:** This is the **hidden boss** of your presentation. It controls the design of all other slides. If you put your school logo or a specific background on the Master Slide, it will appear on **every slide** automatically. Changing the font on the Master changes it everywhere. This keeps your presentation looking unified.
 - *Find it:* **View > Slide Master.**
- **Templates:** These are pre-designed presentations with beautiful backgrounds, fonts, and colour schemes. Using a template gives you a professional look instantly, so you can focus on your content.

2. Multimedia Integration

- To make your presentation engaging, add more than just text!
- **Insert Video:** You can add a video clip from your computer or from the internet (like YouTube). For example, in a presentation about animals, you can show a short clip of a lion roaring.
 - *Go to:* **Insert > Video.**
- **Insert Audio:** You can add background music or a voice recording. For example, you can record yourself explaining a slide and play it during the presentation.
 - *Go to:* **Insert > Audio.**

3. Animations and Transitions (The Difference!)

- This is very important to understand:
 - **Transitions:** This is the **movement or effect when you change from one SLIDE to the next SLIDE**. Examples: **Fade, Push, Zoom, Split**.
 - **Animations:** This is the **movement or effect you apply to an OBJECT (text, picture, shape) ON A SINGLE SLIDE**. Examples: text **Flying In**, a picture **Bouncing**, a shape **Spinning**.
- **Simple Rule: Transitions = between slides. Animations = on a slide.**



4. Hyperlinks and Presenter Tools

- **Hyperlinks:** You can turn any text or picture into a clickable button. During your presentation, clicking it can:
 - **Jump to another slide** in your presentation (e.g., "Click here to see the summary").
 - **Open a website** (e.g., "Visit our school website").
 - *To use:* Select text/picture -> **Insert > Link**.
- **Presenter Notes:** At the bottom of the editing screen, there is a notes section for each slide. You can write reminders or the full script of what you want to say. **During the**

slideshow, only you (the presenter) can see these notes on your screen, while the audience sees only the slide.

EVALUATION

1. If you want your school's name to appear at the bottom of every slide, which special slide should you add it to?
2. Name one type of multimedia file (not a picture) you can add to a slide to make it more interesting.
3. What is the visual effect called when a slide "wipes away" to reveal the next slide: Animation or Transition?
4. You are presenting and you click on a word that says "More Details." It takes you to slide number 10. What is that word called?
5. Where can you write secret notes to remind yourself of what to say for each slide, which the audience will not see?
6. Why is it a good idea to use a template when you start a new presentation?

ASSIGNMENT

1. **(Creative Thinking/Oral):** You are creating a presentation titled "Sounds of Nigeria." Describe **one sound effect or piece of music** you would add to a slide showing a busy Lagos market.
 2. **(Drawing):** On a plain sheet of paper, **draw the layout** for a single presentation slide titled "My Favorite Food: Jollof Rice." Include a spot for a title, a bullet-point list (for ingredients), and a box where a picture would go.
 3. **(Written):** List **two different slide transition effects** you have seen in any presentation (in church, school, or on TV).
-

WEEK 9: ADVANCED SPREADSHEETS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Use simple functions including SUM, MAX, and MIN.
2. Explain the difference between absolute and relative cell referencing.
3. Create a simple chart to represent data.
4. Sort and filter data in a spreadsheet.

CONTENT

1. Complex Formulas and Functions

- **Functions** are pre-built formulas that do specific calculations for you.
 - **=SUM(A1:A10)** – **Adds up** all numbers from cell A1 to A10.
 - **=MAX(B2:B20)** – Finds the **largest** (maximum) number in that range.
 - **=MIN(B2:B20)** – Finds the **smallest** (minimum) number in that range.
 - **=COUNT(C1:C100)** – **Counts how many cells** in that range contain numbers.
 - **=IF Statement:** This makes a decision. **=IF(D5>=50, "PASS", "FAIL")** means:
If the score in cell D5 is 50 or more, then write "PASS", otherwise write "FAIL".

2. Cell Referencing: Relative vs. Absolute

- This is about what happens when you **copy** a formula from one cell to another.
- **Relative Reference (Default):** The cell addresses in the formula **change automatically** when copied. If you copy **=A1+B1** from row 1 to row 2, it becomes **=A2+B2**. It's "relative" to its new position.
- **Absolute Reference (Using \$):** The cell address **stays locked/fixed** when copied. You use the dollar sign (\$). If you have a tax rate in cell **C\$1** and copy a formula down, the reference to **C\$1** will stay the same for every row. **\$A\$1** locks both column and row.

A	B	B	D
Price	Quantity	Tax Rate	Total
	2	$=A2*B2 +$	
100	2	$(A2*B2*C1)$	
200	$=A3*B3 + (A3*B3*\$C\$1)$		
200	2 error	$\$C\1	
3	3		660

3. Charts and Graphs (Turning Numbers into Pictures)

- Charts help people understand data quickly.
- **Bar Chart:** Uses bars of different lengths. Best for **comparing different items** (e.g., comparing the population of 5 different Nigerian states).
- **Pie Chart:** Looks like a sliced pie. Best for showing **parts of a whole** or percentages (e.g., how you spend your weekly allowance: 50% food, 30% savings, 20% transport).
- **Line Chart:** Uses points connected by lines. Best for showing **changes over time** (e.g., temperature each day of the week).

4. Sorting and Filtering (Organizing Your Data)

- **Sorting:** Arranging your data in a specific order.
 - **A to Z / Smallest to Largest = Ascending** order.
 - **Z to A / Largest to Smallest = Descending** order.
 - *Example:* Sort a class list by last name alphabetically.
- **Filtering:** This **hides** rows of data you don't want to see right now, showing only rows that match your criteria. It does not delete them.

- *Example:* In a list of all students, filter to show **only** pupils in "Primary 6" or **only** students who scored "A" in Maths.

EVALUATION

1. Your teacher has test scores in cells F5 to F30. What function will instantly find the highest score?
2. What does the =COUNT function do?
3. You have a list of 50 products and their prices. You want to see only items that cost more than ₦1000. Should you use **Sorting** or **Filtering**?
4. You want to show how your class of 30 pupils voted for Class Captain (3 candidates). Which chart—Bar Chart or Pie Chart—is best to show the result? Why?
5. In the cell reference \$D\$5, what does the \$ symbol tell the spreadsheet to do?
6. Is arranging names from Z to A an example of ascending or descending order?

ASSIGNMENT

1. **(Practical/Written):** Write down these five numbers: 15, 8, 22, 17, 3. **Circle** the **MAX** (highest) number. **Underline** the **MIN** (lowest) number.
 2. **(Drawing):** Draw a simple **Pie Chart**. Divide the circle into three sections and label them: **Sleep (8 hrs)**, **School (7 hrs)**, **Play (9 hrs)**. (You don't need exact maths, just a sketch).
 3. **(Applied Thinking):** Give **one reason** why the owner of a small shop (like a provision store) would find a spreadsheet useful.
-

WEEK 10: DATABASE BASICS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a database.
2. Identify the three main structures of a database (Table, Record, Field).
3. Differentiate between a database and a spreadsheet.
4. Explain the purpose of a Query.

CONTENT

1. Introduction to Databases

- A **Database** is an **organized, electronic collection of information (data)** that can be easily stored, searched, and managed.
- **Real-Life Analogy:** A **paper filing cabinet** in an office. Each drawer has folders, and each folder has information about one person or thing. A digital database is a super-powered, fast, electronic version of that cabinet.
- **Examples:** Your school's register of all students, a hospital's patient records, a library's book catalog, or a phone's contact list.

2. Database Structure: Tables, Records, and Fields

All databases are built on three main parts:

1. **Table:** The container that holds all the related information. It looks like a grid or spreadsheet. *Example:* A "Pupils" table.
2. **Field (Column):** A single **category** or type of information. Each column is a field.
 - *In the "Pupils" table, fields could be: **PupilID, FirstName, LastName, Age, Class, ParentPhone.***
3. **Record (Row):** A complete set of information about **one** specific person or item. A row is a record.
 - *One record in the "Pupils" table could be: P001 | Chiamaka | Nwachukwu | 11 | Primary 6 | 0803xxx1234*

TABLE

RECORDS (ROWS)

3. Databases vs. Spreadsheets

- **Spreadsheets (like Excel):**
 - **Best for:** Calculations, formulas, and creating charts/graphs.
 - **Like a:** Super calculator and artist for numbers.
- **Databases (like Access):**
 - **Best for:** Storing **very large amounts** of data (thousands of records) and finding, filtering, and reporting on specific pieces of that data very quickly.
 - **Like a:** Powerful, organized library where you can ask complex questions.
- **Simple Difference:** You use a spreadsheet to **calculate your monthly budget**. A bank uses a database to **store account details for all its millions of customers**.

4. Queries and Forms

- **Query:** This is the most powerful tool. A query is a **question** you ask the database. The database searches through all records and gives you the exact answer.
 - *Example Query:* "Show me all pupils in **Primary 6** who are **12 years old**." The database finds and displays only those records.
- **Form:** A form is a **user-friendly window** for entering or viewing data one record at a time. It's easier and less confusing than typing directly into the big table. It looks like a digital form you fill online.

EVALUATION

1. What is a simple definition of a database?
2. In a "School Library" database table, what is one complete row of information about a single book called?
3. In the same library table, what do we call the column that holds the "Author's Name" for all books?
4. Your father wants to track his daily business expenses and make a monthly profit chart. Is a **Spreadsheet** or a **Database** the better tool for him? Why?
5. What is the name of the database tool that lets you ask a question like "Which students scored above 80%"?
6. Give one example of a database used in everyday life in Nigeria.

ASSIGNMENT

1. **(Drawing):** Design a simple "My Class Library" database table. **Draw the grid** with 3 columns (Fields): **Book Title, Author, Borrower's Name**. Fill in 2 rows (Records) with pretend book data.
 2. **(Analysis):** On your drawing from Assignment 1, **draw a circle** around one **Field** name.
 3. **(Creative Thinking/Written):** You are the Head Teacher. Write **one question (Query)** you could ask the school's pupil database to get useful information. (e.g., "How many pupils live in the Victoria Island area?").
-

WEEK 11: REVISION

CONTENT

1. Review of Weeks 1-4 (Systems, Networks, Internet, Cloud):

- Recap: How computers use binary (0/1). Bits vs. Bytes. The IPO Cycle (Input, Process, Output, Storage). Factors that make a computer fast (CPU, RAM, SSD).
- Recap: What is a network? Difference between LAN (small area) and WAN (large area, like the Internet). Key devices: Modem, Router, Switch. Benefits of networking.
- Recap: Internet (the road) vs. World Wide Web (the shops on the road). Parts of a URL. What a browser does. Smart searching tips.
- Recap: "The Cloud" as online storage. Examples: Google Drive, OneDrive. Benefits: Access anywhere, safe backup. Importance of strong passwords.

2. Review of Weeks 7-10 (Software Applications):

- Recap (Word): Using Styles, Columns, Mail Merge, Track Changes/Comments, and automatic Table of Contents.
- Recap (Presentations): Master Slide vs. regular slides. Adding video/audio. Difference between Transitions (between slides) and Animations (on a slide). Using Hyperlinks.
- Recap (Spreadsheets): Key functions: SUM, MAX, MIN, COUNT, IF. Relative vs. Absolute (\$) reference. Creating Bar and Pie charts. Sorting and Filtering data.
- Recap (Databases): Definition. Structure: Table, Record (row), Field (column). Difference from a spreadsheet. Purpose of a Query (to ask questions).

3. Completing pending notes and practical exercises.

- The teacher will go through any unfinished topics.
- Pupils will complete and correct all previous classwork and assignments.
- Interactive Q&A session to address any remaining difficulties.

PRIMARY 6 BASIC DIGITAL LITERACY SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Cybersecurity Fundamentals	Understanding cybersecurity: protecting systems and data. Common threats: viruses, malware, hackers, phishing, ransomware. Antivirus software importance. Strong passwords: complexity, uniqueness, password managers. Two-factor authentication. Software updates importance.
2	Social Media Safety	Social media platforms overview. Age restrictions and why they exist. Privacy settings mastery. Appropriate posting guidelines. Oversharing risks. Managing friend requests. Dealing with negativity online. Balance and time management.
3	Online Reputation Management	Digital reputation importance. Googling yourself. Managing privacy across platforms. Professional online presence basics. Impact on future opportunities. Fixing mistakes online. Positive online contribution. Building digital portfolio.
4	Critical Evaluation of Online Information	Fake news and misinformation. Evaluating sources: authority, accuracy, objectivity, currency. Fact-checking techniques. Lateral reading. Understanding bias. Media literacy. Responsible sharing.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Text-Based Programming Introduction	Transition from block to text-based coding. Introduction to Python: beginner-friendly language. Python environment setup. Basic syntax: print statements, comments. Variables: storing data,

		naming conventions. Data types: strings, integers, floats.
8	Python Programming I	User input: input() function. Arithmetic operations. String operations: concatenation, formatting. Type conversion. Simple calculator program. Interactive programs. Errors and debugging.
9	Python Programming II	Conditional statements: if, elif, else. Comparison and logical operators. Making decisions in programs. Creating interactive stories or quizzes. Nested conditionals. Program flow control.
10	Python Programming III	Loops in Python: for loops, while loops. Range function. Iterating through data. Loop control: break, continue. Creating patterns and games. Combining conditionals and loops.
11	REVISION	Review of all topics. Complete Python projects. Code presentations and peer review.
12	EXAMINATION	End of Second Term Examination (Theory and Practical)
13	CLOSING	Report distribution and vacation

PRIMARY 6 DIGITAL LITERACY LESSON NOTES

SECOND TERM

WEEK 1: FABRIC DECORATION BY TIE AND DYE METHODS I

OBJECTIVES By the end of the lesson, pupils should be able to:

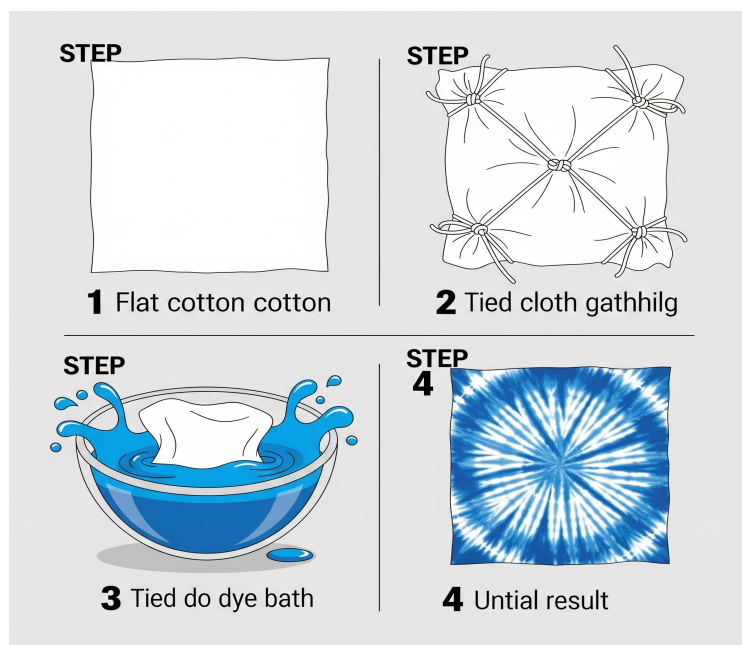
1. Define Tie and Dye in their own words.
2. Mention the local Nigerian name for tie and dye (Adire) and its cultural importance.
3. List and describe the use of four main materials needed for tie and dye.
4. Identify the three primary colours and demonstrate how to mix them to get secondary colours.

CONTENT

1. Meaning and History of Tie and Dye

Tie and Dye is a fun way of creating patterns on cloth by **preventing colour from reaching some parts**. You do this by tying, folding, or stitching the fabric very tightly before dipping it in dye. The tied parts stay white (or the original colour), while the loose parts soak up the dye, making beautiful designs.

- **Cultural Significance in Nigeria:** In Nigeria, especially among the Yoruba people, this art is famously called **Adire** (pronounced Ah-dee-reh). Adire cloth is more than just fabric; it is part of our culture. People wear specially designed Adire for important events like weddings, festivals (e.g., Eyo, New Yam festival), and naming ceremonies. It is a way of showing our beautiful heritage and skills.



2. Materials Needed for Tie and Dye

You need simple, easy-to-find items:

- **Fabric: White 100% cotton cloth** is the best. Cotton is like a sponge; it sucks up the dye very well. An old white handkerchief, T-shirt, or pillowcase works perfectly.
- **Dye:** This is the coloured powder or liquid that gives the cloth its colour. We often use **reactive dyes** or **indigo** for Adire.
- **Resist Materials:** These are used to tie the cloth and block the dye. You can use:
 - Raffia (from palm trees)
 - Strong thread or twine
 - Rubber bands (very easy for beginners)
 - Small stones or seeds (wrapped inside the cloth)
- **Tools and Safety Gear:**
 - Plastic buckets or bowls (for mixing dye)
 - Hot water (usually handled by the teacher)
 - Stirring stick (like a wooden spoon)
 - **Plastic gloves** to protect your hands from stains.
 - **An apron or old clothes** to protect your uniform.

3. Safety Precautions (VERY IMPORTANT)

Working with dyes and hot water requires care.

1. **Always wear your gloves.** Dye can stain your skin for days.
2. **Do not put your face close to the dye powder.** It can be harmful if you breathe it in.
3. **Never touch or play with the hot water** used for mixing dye. Let your teacher handle it.
4. Work in a well-ventilated area or outside.

4. Basic Tying Techniques

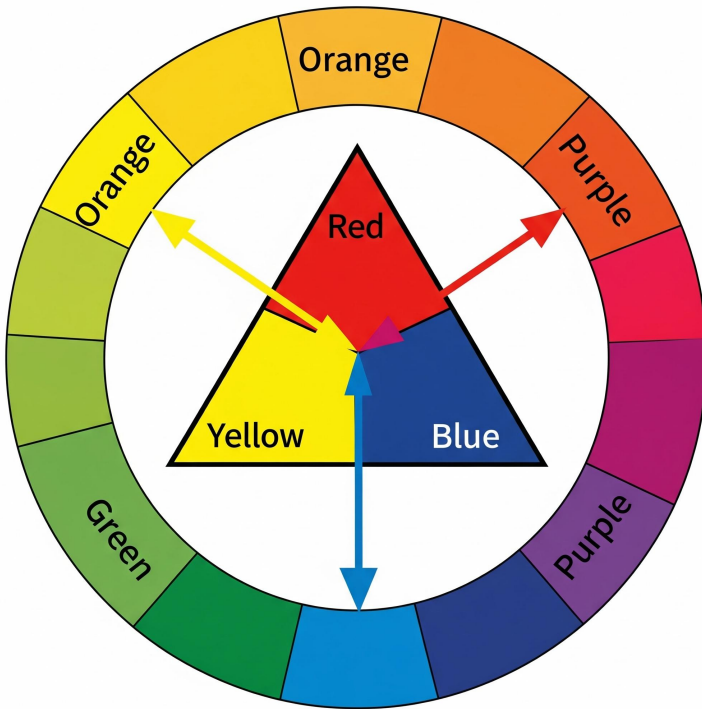
There are many ways to tie. We will start with three simple ones:

- **Knotting:** Simply make a tight knot on a section of the cloth.
- **Pebbling:** Place a small stone or bean on the cloth, pull the fabric around it, and tie it tightly with string. This creates a circle.
- **Stripes:** Fold the cloth like a fan (accordion fold), then tie strings or rubber bands at different points along the fold.

5. Colour Theory: Creating All Colours

Artists know that just a few colours can make many others.

- **Primary Colours:** These are the **boss colours**. They cannot be created by mixing other colours. They are: **RED, YELLOW, and BLUE.**
- **Secondary Colours:** You make these by mixing two primary colours equally.
 - Red + Yellow = **ORANGE**
 - Yellow + Blue = **GREEN**
 - Blue + Red = **PURPLE (or VIOLET)**



EVALUATION

1. What is the main idea behind tie and dye? (To stop dye from reaching some parts of the cloth).
2. What is the Yoruba name for this art, and when do people wear it?
3. Why is a white cotton cloth better than a coloured nylon cloth for tie and dye?
4. Name two safe materials you can use to tie your cloth with.
5. If you mix yellow dye and blue dye, what new colour will you create?
6. Why must you wear gloves during the dyeing process?

ASSIGNMENT

1. **(Practical):** Bring a small, white cotton cloth (e.g., an old handkerchief) to the next class for our practical session.
2. **(Drawing):** Draw and colour a simple colour wheel. Label the three primary and three secondary colours.

3. **(Oral/Listing):** Look around your home and tell your parent three items you see that could be used as 'resist materials' for tying cloth (e.g., rubber band, strong thread from a sewing box).



WEEK 2: FABRIC DECORATION BY TIE AND DYE METHODS II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Demonstrate and describe how to create "Spiral" and "Bull's Eye" patterns.
2. Explain the key difference between Tie-Dye and Batik.
3. Describe the step-by-step process of dyeing fabric.
4. Create a simple tied pattern on their own piece of fabric.

CONTENT

1. Advanced Tying Techniques

Now, let's learn some popular patterns:

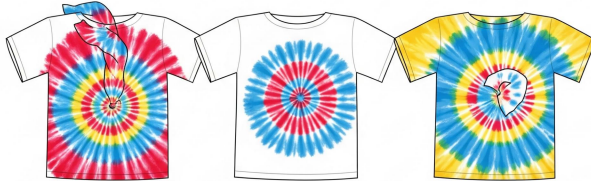
- **Spiral (Swirl Pattern):**
 - a. Lay your cloth flat on a table.
 - b. Pinch the very centre of the cloth with two fingers.
 - c. Start twisting the cloth around that centre point. Keep twisting until the whole cloth looks like a flat, round snail shell or a **swirled lollipop**.
 - d. Use 3 or 4 rubber bands to hold the shape, wrapping them around the spiral like slices of a pie.
- **Bull's Eye (Circle/Target Pattern):**
 - a. Pinch a point on the cloth and pull it up, like you are making a little fabric mountain.
 - b. Wrap a rubber band very tightly around the base of this "mountain". This will be the centre of your bull's eye.
 - c. Move down a little and wrap another rubber band. You can add more bands to create multiple circles.
- **Marbling (Crumple Technique):**
 - a. Simply crumple the entire cloth into a loose, random ball.

- b. Tie a few rubber bands around it in different directions. This creates random, stone-like patterns.

Spiral Technique



Bull's Eye



2. Introduction to Batik

Tie-Dye is not the only way to make patterns on cloth. **Batik** is another important method.

- **Main Difference:** In Batik, melted **wax** (not string) is used to block the dye. The artist uses a tool called a *tjanting* to draw with hot wax on the cloth. When the cloth is dyed, the waxed areas resist the colour. The wax is later removed with hot water or ironing.

3. The Dyeing Process: Step-by-Step

Here is what we will do in our practical class:

1. **Prepare the Fabric:** Wash your white cloth to remove any dirt or starch. This helps the dye soak in evenly.
2. **Tie Your Design:** Use your chosen technique (spiral, bull's eye, etc.) and tie it **very tightly**. If it's loose, dye will sneak in and spoil the pattern.
3. **Mix the Dye Bath:** (Teacher will do this). The dye powder is mixed with hot water and a fixative chemical (like Hydrosulphite) so the colour stays on the cloth.

4. **Dye the Cloth:** Wearing gloves, gently place your tied cloth into the dye bath. Push it down so it's fully covered. Leave it for 15-20 minutes.
5. **Rinse:** Take the cloth out and rinse it in a bucket of cold water until the water runs clear (not coloured). This removes excess dye.
6. **Untie and Dry:** Carefully cut the rubber bands or strings. Be amazed at your pattern! Open the cloth and hang it to dry in a **shady place**. Direct sun can fade the bright colours.

EVALUATION

1. To make a spiral pattern, what do you do after pinching the centre of the cloth?
2. What is used to resist the dye in Batik: wax or rubber bands?
3. Why is the first step to wash the fabric before dyeing it?
4. What should you do with the dyed cloth immediately after taking it out of the dye bath?
5. Why is it important to tie the string or rubber band very, very tight?

ASSIGNMENT

1. **(Practical):** Use the white cloth you brought. Practice creating a "Spiral" pattern on it using rubber bands. (We will dye it next week).
2. **(Listing):** List three useful items that can be made from tie-dyed fabric (e.g., head tie, pillow cover, tote bag).
3. **(Written):** In one sentence, explain why we rinse the fabric in cold water after dyeing.

WEEK 3: FABRIC DECORATION BY TIE AND DYE METHODS III

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the importance of planning a design before starting.
2. Differentiate between Natural dyes and Synthetic dyes with examples.
3. Discuss simple ways to market and sell tie and dye products.
4. Display and discuss their finished tie and dye project.

CONTENT

1. Design Planning: Think Before You Tie!

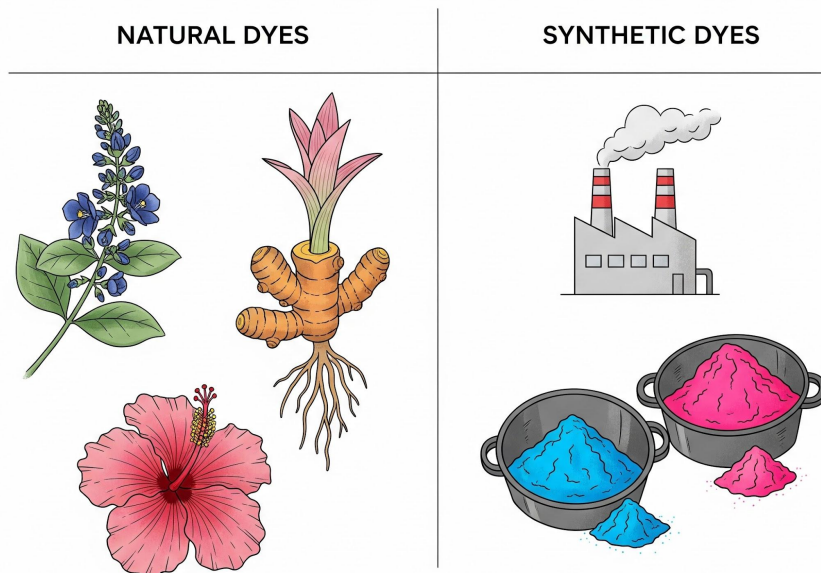
A good artist plans. Before you tie your cloth, it is helpful to **sketch your idea** on paper.

- Ask yourself: Do I want circles (bull's eye)? Swirls (spiral)? Stripes?
- Which colours will I use? One colour? Two colours?
- Planning helps you avoid a messy, muddy-looking cloth and creates a beautiful, intentional design.

2. Natural Dyes vs. Synthetic Dyes

Dyes can come from nature or from a factory.

- **Natural Dyes:** Made from plants, fruits, vegetables, and even insects!
 - **Examples:** Turmeric or onion skins (yellow/brown), indigo leaves (blue), hibiscus flowers (red/pink), avocado pits (pink).
 - **Advantages:** They are environmentally friendly and safe.
 - **Disadvantages:** Colours are often softer and can fade faster with washing and sun.
- **Synthetic Dyes:** Made from chemicals in a laboratory.
 - **Examples:** The bright dye powders we often use in class (e.g., Procion MX dyes).
 - **Advantages:** They give very **bright, strong colours** that last a long time.
 - **Disadvantages:** The waste water can harm rivers and soil if not disposed of properly.



3. Entrepreneurship: Turning Art into Business

Tie and Dye is not just art; it can be a **small business** (entrepreneurship). You can create and sell items to make money.

- **Products to Sell:** Trendy backpacks, customised school T-shirts for clubs, table runners, wrappers (iro), and shirts.
- **How to Market (Advertise):**
 - Show your best work to friends and family.
 - Display items at school events or local markets.
 - Take clear, good pictures of your products.
- **Pricing:** The price should cover the cost of your cloth, dye, and time, plus a little extra as profit.

4. Project Completion and Display

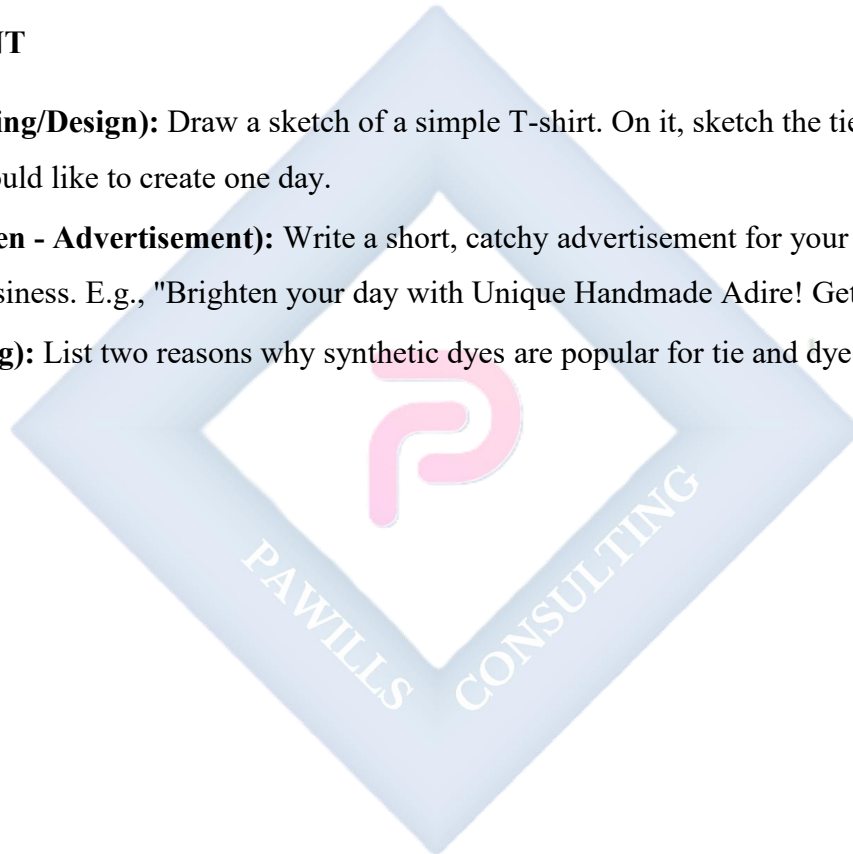
Today, we will complete our dyeing process! After untying and drying your cloth, you can iron it to make it smooth and neat. We will then have a **class exhibition** where everyone displays their work. Look at your friends' designs—see how creative they are!

EVALUATION

1. Why is it a good idea to draw your design on paper first?
2. Name one plant in Nigeria that can be used to make a natural dye.
3. Which type of dye gives the brightest, longest-lasting colours: natural or synthetic?
4. If you made a beautiful tie-dye head tie, who are two groups of people you could try to sell it to?
5. Why shouldn't we pour leftover synthetic dye water down the kitchen sink?

ASSIGNMENT

1. **(Drawing/Design):** Draw a sketch of a simple T-shirt. On it, sketch the tie-dye pattern you would like to create one day.
2. **(Written - Advertisement):** Write a short, catchy advertisement for your imaginary tie-dye business. E.g., "Brighten your day with Unique Handmade Adire! Get yours now!"
3. **(Listing):** List two reasons why synthetic dyes are popular for tie and dye projects.



WEEK 4: CHORAL SINGING I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define choral singing and contrast it with solo singing.
2. List and describe the four main voice parts in a choir (SATB).
3. Demonstrate the correct standing and breathing posture for singing.
4. Perform simple breathing exercises to support the voice.

CONTENT

1. What is Choral Singing?

Choral singing is when a group of singers, called a **choir**, performs music together. The key word is **together**. It is not about one person being the loudest; it is about blending voices to create one beautiful, harmonious sound. You must learn to listen to others while you sing.

2. Types of Choirs

Choirs can be formed in different ways:

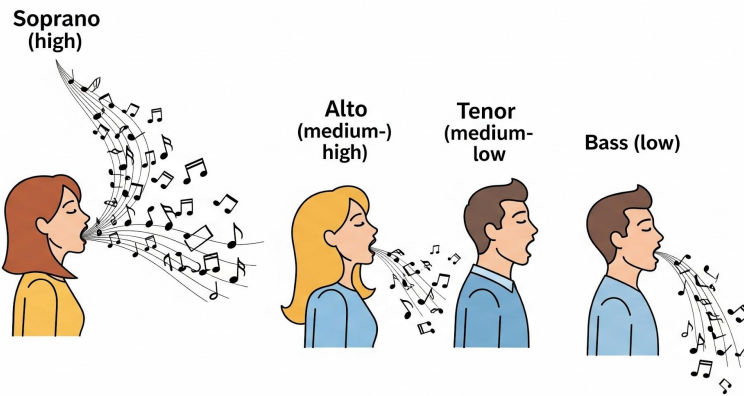
- **Children's Choir:** Made up of boys and girls. Their voices are often called **treble** voices.
- **Mixed Choir:** The most common type. It has both male and female singers.
- **Male Voice Choir:** Only men and boys.
- **Female Voice Choir:** Only women and girls.

3. Voice Parts in a Mixed Choir (SATB)

In a mixed choir, voices are grouped into four parts based on their **pitch** (how high or low they sing). We use the short form **SATB**.

- **Soprano (S):** The highest female voice. They often sing the **melody** (the main tune you remember).
- **Alto (A):** The lower female voice. They sing the **harmony** that supports and enriches the melody.
- **Tenor (T):** The higher male voice. They also sing harmony, usually above the basses.

- **Bass (B):** The lowest male voice. They sing the foundation notes, like the deep roots of a tree.



4. Posture and Breathing: The Singer's Foundation

To sing well, you need to stand well and breathe correctly.

- **Correct Singing Posture:**
 - a. Stand with your feet shoulder-width apart, one slightly in front of the other for balance.
 - b. Knees should be relaxed, not locked.
 - c. Stand tall, like a string is pulling your head up. Shoulders should be down and relaxed.
 - d. Look forward. Imagine you are singing to someone at the back of the room.
- **Breathing from the Diaphragm:** Good singers don't just breathe with their chests. They use their **diaphragm** (a muscle below your lungs).
 - **Exercise:** Place your hands on your tummy. Breathe in deeply through your nose and feel your tummy push your hands out. Breathe out slowly with a "ssss" sound. Your shoulders should not go up and down.

EVALUATION

1. What is the main difference between singing in a choir and singing alone?
2. What does "SATB" stand for?
3. Which voice part usually sings the highest notes in a mixed choir?
4. Which voice part sings the lowest, foundational notes?
5. Show me the correct standing posture for singing.
6. Where should you feel your breath moving when you breathe correctly for singing?
(Your tummy).

ASSIGNMENT

1. **(Practical):** Stand in front of a mirror at home. Practice your singing posture for one minute. Say your name out loud while maintaining the posture.
2. **(Listening/Oral):** Listen to any choral music (e.g., a church choir on the radio). Try to identify if you can hear high voices (sopranos) and low voices (basses).
3. **(Written):** Write down the four voice parts of a choir and write one sentence to describe the Alto voice.

WEEK 7: CHORAL SINGING II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define harmony and differentiate it from melody.
2. Explain the terms Dynamics, Crescendo, and Decrescendo.
3. Demonstrate clear diction (pronunciation) while singing a familiar song.
4. Follow simple basic hand signals from a conductor.

CONTENT

1. Harmony: Singing Together Beautifully

Harmony is the combination of different musical notes sung or played at the same time. It is what makes choir music sound rich and full.

- **Melody:** This is the main tune, the part you whistle. In a choir, the Sopranos often sing this.
- **Harmony:** These are the other notes that the Altos, Tenors, and Basses sing. They fit under, over, and around the melody like a musical puzzle, making it sound wonderful.

2. Dynamics: The Volume of Music

Dynamics tell us how **loud** (forte) or **soft** (piano) to sing. Changing dynamics makes music exciting!

- **Forte (f):** Means "loud." Sing with strong, full voices.
- **Piano (p):** Means "soft." Sing gently and quietly.
- **Crescendo (<):** A gradual increase in volume. Start soft and get louder, like an approaching train.
- **Decrescendo (>):** A gradual decrease in volume. Start loud and get softer, like a train moving away.

3. Diction and Articulation: Singing Clearly

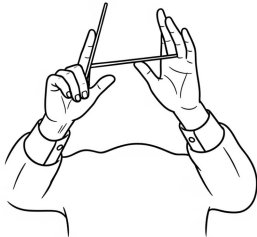
Diction means pronouncing words clearly so the audience can understand the song's story or message.

- Open your mouth wider than when you talk.
- Pay special attention to the **end of words** (like "sing" not "sin").
- **Articulation** is about making consonants crisp (T, D, P, B, K). For example, sing "Praise" with a strong "P" sound.

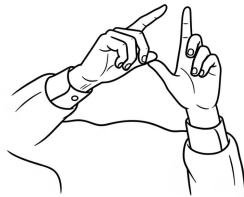
4. The Conductor: The Leader of the Choir

The **conductor** uses hand gestures and facial expressions to lead the choir.

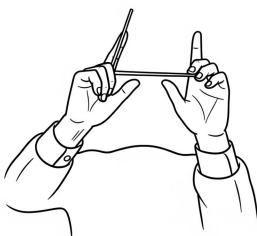
- **The Prep Beat:** The conductor raises their hands to tell you to get ready and breathe in.
- **The Downbeat:** The first downward motion tells you when to start singing.
- **Cut-Off:** A closing or pulling-in motion of the hand tells you when to stop.
- **Dynamics:** Hands open wide for loud (forte), hands close together for soft (piano).



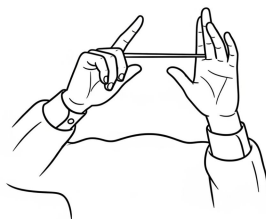
Music



Conductor Signals



Conductor



Condrsignals

EVALUATION

1. If the Altos are singing different notes from the Sopranos but it sounds good, what are they creating?

2. What is the musical word for getting gradually louder?
3. Why is it important for a choir to have good diction?
4. Who uses hand signals to lead the choir's volume and timing?
5. Sing the first line of "Nigeria, We Hail Thee" first piano (softly), then forte (loudly).

ASSIGNMENT

1. **(Written):** Write the definitions for: 1. **Piano** (in music). 2. **Forte** (in music).
2. **(Practical):** Practice singing the first stanza of the National Anthem. Focus on opening your mouth wide and pronouncing every word clearly. Record yourself if possible.
3. **(Drawing):** Draw a simple sketch of a conductor showing the choir to sing softly.



WEEK 8: CHORAL SINGING III

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define "A Cappella" singing.
2. List three important rules of choir etiquette and discipline.
3. Suggest two ways to overcome stage fright.
4. Describe how to hold and use a microphone properly.

CONTENT



1. Repertoire and Stage Presence

- **Repertoire:** This is the collection of songs a choir has learned and is ready to perform. A good choir has a varied repertoire—fast songs, slow songs, songs in different languages.
- **Stage Presence:** This is how you look and act while performing. It includes:
 - **Smiling** (when appropriate).
 - **Standing still** and not fidgeting.
 - **Making eye contact** with the audience or the conductor.
 - Looking confident and enjoying the music.

2. A Cappella Singing

A Cappella (pronounced ah-kah-pel-ah) is singing **without any instrumental accompaniment**—no piano, no drums, no guitar. The choir provides all the music and rhythm with their voices alone. This requires excellent listening skills and teamwork to stay perfectly in tune and in time.

3. Choir Discipline and Etiquette

A choir is a team. Good discipline makes rehearsals productive and performances successful.

1. **Be Punctual:** Arrive early for practice. Latecomers disturb everyone.
2. **Be Silent:** Do not talk when the conductor or another section is working. Listen.
3. **Pay Attention:** Keep your eyes on the conductor. Follow instructions quickly.
4. **Be Prepared:** Bring your music (if you have it) and a pencil to mark instructions.
5. **Be Respectful:** Respect your conductor, the pianist, and fellow choir members.

4. Managing Stage Fright and Using a Microphone

- **Stage Fright:** It is normal to feel nervous before performing.
 - **How to cope:** Take slow, deep breaths. Be very well-prepared (know your parts). Focus on sharing the message of the song with the audience, not on your fear.
- **Microphone Technique:** If you use a mic, do it correctly.
 - Hold the mic by its **handle**. Never cover the head (top) with your hand.
 - Hold it about a **fist's distance** from your mouth.
 - Speak or sing **normally**; do not shout. The microphone will make you loud.
 - If the mic is on a stand, adjust the stand to your height before you start.

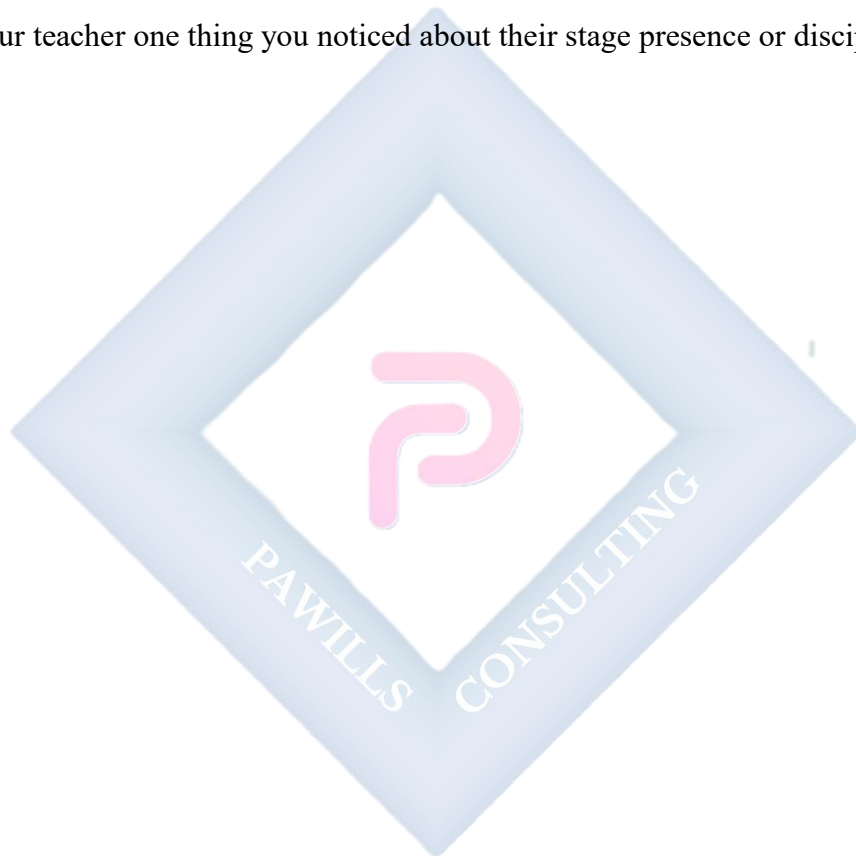
EVALUATION

1. What does "A Cappella" mean?
2. What is one important rule about timekeeping for a choir member?
3. Why should choir members keep their eyes on the conductor during a performance?
4. What is one thing you can do backstage to calm your nerves before singing?

5. If you are singing with a handheld microphone, how far should it be from your mouth?

ASSIGNMENT

1. **(Listing):** List three songs you think would be good for your class choir to learn and sing. They can be folk songs, patriotic songs, or school songs.
2. **(Written):** Write a short paragraph (3-4 sentences) on why discipline and teamwork are important for a choir's success.
3. **(Observation/Oral):** If you can, watch a video of a choir performance (online or on TV). Tell your teacher one thing you noticed about their stage presence or discipline.



WEEK 9: THEATRICAL ELEMENTS AND PERSONNEL

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List and describe four key elements of drama/theatre.
2. Identify different areas on a stage (Upstage, Downstage, Center Stage).
3. Differentiate between the roles of a Director and a Producer.
4. Mention the role of a Costume Designer and a Set Designer.

CONTENT

1. Key Elements of Theatre (Drama)

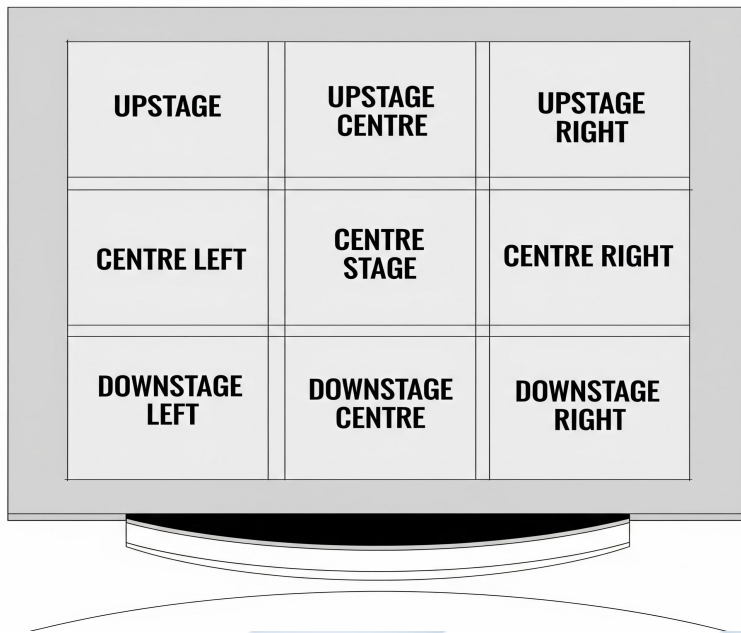
A good play is built on these important parts:

- **Plot:** The **storyline**. What happens first, next, and last? It is the sequence of events.
- **Character:** The **people (or animals/things)** in the story. They have personalities, feelings, and goals.
- **Dialogue:** The **words spoken** by the characters. This is how they talk to each other and tell the story.
- **Spectacle:** Everything the audience **sees**—the visual parts. This includes costumes, makeup, sets, lighting, and special effects. It makes the play exciting to watch!

2. Stage Areas: Knowing Where to Go

The stage is divided into areas so the director can tell actors where to move. Imagine the stage as a big square.

- **Center Stage:** The very middle. This is a strong, important spot.
- **Downstage:** The area **closest to the audience**. When you are here, the audience can see your face clearly.
- **Upstage:** The area **farthest from the audience** (towards the back wall).
- **Stage Left & Stage Right:** These are from the **actor's point of view** when they are facing the audience. *Your* left is **Stage Left**.



3. Theatre Personnel: The Production Team

Putting on a play is a team effort. Many people have special jobs:

- **Playwright:** The writer. They create the script (the written play).
- **Director:** The **creative leader**. They interpret the script, guide the actors on how to speak and move, and work with all designers to create the overall vision.
- **Producer:** The **business and money manager**. They find the money to pay for the play, rent the theatre, and handle tickets and advertising.
- **Stage Manager:** The **organizer during the show**. They call for lights and sound cues, make sure actors are ready, and solve any backstage problems.
- **Costume Designer:** Decides what each character will wear to show their personality, job, and the time period of the play.
- **Set Designer:** Designs and builds the **scenery and background**—the rooms, forests, or streets where the action happens.

EVALUATION

1. Which element of theatre refers to the order of events in the story?
2. If your teacher tells you to move "Downstage," should you move towards the audience or away from them?
3. Who is the person who writes the script for the play?
4. What is the main job of the Costume Designer?
5. Who is in charge of the creative side of the play, telling actors how to perform?

ASSIGNMENT

1. **(Drawing):** Draw a simple diagram of a stage and label these three areas: **Center Stage, Downstage, Upstage.**
2. **(Written Reflection):** If your class were to perform a play, which backstage role (e.g., Director, Costume Designer, Stage Manager) would you be most interested in and why? (2-3 sentences).
3. **(Practical/Writing):** Write a short, 4-line dialogue between two characters. One is a buyer at a market, the other is a seller arguing about the price of bananas.

WEEK 10: CARE OF TOOLS AND EQUIPMENT IN PERFORMING ARTS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List three different categories of equipment used in performing arts.
2. Explain how to properly care for and store costumes.
3. State two important safety rules for handling electrical equipment (lights, microphones).
4. Discuss the importance of respecting shared tools and resources.

CONTENT



1. Types of Performing Arts Equipment

Performing arts use many special tools:

- **Musical Instruments:** Drums (like the conga or talking drum), keyboards, recorders, guitars.
- **Costumes:** The special clothing worn by actors and dancers for performances.
- **Props (Properties):** Objects that actors use on stage—a king's crown, a letter, a basket, a fake sword.

- **Technical Equipment:** This makes the show look and sound good. Includes microphones, speakers, amplifiers, stage lights, and their cables.

2. Care and Maintenance of Equipment

Taking care of equipment shows you are a responsible performer and saves the school money.

- **Musical Instruments:**
 - Handle them gently. Do not bang drums unnecessarily.
 - Keep them in their cases or covered when not in use to protect from dust and damage.
 - Report any broken strings or parts to the teacher immediately.
- **Costumes:**
 - Handle them carefully to avoid tears.
 - After use, they should be **hung up properly** on hangers, not thrown on the floor. This prevents wrinkles and keeps them neat for the next performer.
 - Some costumes may need to be washed or dry-cleaned.
- **Props:**
 - Use props only for their intended purpose during the play.
 - After the scene, return them immediately to the **props table** or box. Do not take them home or play with them as toys.

3. Safety with Electrical Equipment

Electricity can be dangerous if not respected.

1. **Never touch plugs, sockets, or cables with wet hands.**
2. **Do not pull a plug out by yanking the cable.** Always pull by the sturdy plug head.
3. Make sure cables are taped down or out of walkways so no one trips over them.
4. Turn off amplifiers and mixers when not in use to prevent overheating.

4. Responsibility for Shared Resources

School equipment is for **everyone** to use. If you break a microphone because you were playing with it, the next class cannot use it for their performance. Taking care of shared tools is part of being a good member of the school community. It ensures that everyone gets a chance to create and perform.

EVALUATION

1. What is the difference between a *costume* and a *prop*? Give one example of each.
2. How should you store a fragile instrument like a recorder after using it?
3. Why is it important to hang up costumes instead of folding them in a bag?
4. What is the safe way to unplug an electrical device?
5. Why should we all take care of school performance equipment even if we didn't personally buy it?

ASSIGNMENT

1. **(Practical/Planning):** Create a short "Closing Checklist" for the drama club. List three things that must be done after every rehearsal (e.g., 1. Hang all costumes. 2. Put props in the box. 3. Turn off and unplug the amplifier.)
2. **(Drawing):** Draw your favourite musical instrument. Around it, draw its case or show how it should be stored safely.
3. **(Written - Rule Making):** Write one clear rule for the "Costume Room" that would help keep the costumes in good condition for everyone.

WEEK 11: REVISION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Recall key terms, definitions, and processes from Tie-Dye, Choral Singing, and Theatre.
2. Demonstrate a practical skill learned this term (e.g., explaining a tying technique or demonstrating singing posture).
3. Answer application-based questions confidently in preparation for the terminal examination.

CONTENT

- **Review of Tie and Dye (Adire):**

- Meaning and cultural importance.
- Materials and **safety precautions**.
- Techniques: Spiral, Bull's Eye, Marbling.
- The dyeing process steps (Tie > Dye > Rinse > Untie > Dry).
- Natural vs. Synthetic dyes.
- Entrepreneurship ideas.

- **Review of Choral Singing:**

- Definition and types of choirs.
- Voice parts (**SATB**).
- Correct posture and diaphragm breathing.
- **Harmony** vs. Melody.
- **Dynamics** (Forte, Piano, Crescendo, Decrescendo).
- The role of the Conductor.
- Choir discipline and stage fright management.

- **Review of Theatre (Drama):**

- Key elements: Plot, Character, Dialogue, Spectacle.
- **Stage areas** (Upstage, Downstage, Centre Stage).

- Key personnel: Playwright, Director, Producer, Stage Manager, Designers.
- **Review of Tool Care:**
 - Care for costumes, props, instruments, and electrical gear.
 - Importance of responsibility for shared resources.
- **Practical Revision Session:** The teacher will lead quick, interactive activities:
 - a. *Tie-Dye Quiz:* "Show me with your hands how you would create a bull's eye pattern."
 - b. *Singing Drill:* Practice breathing from the diaphragm together.
 - c. *Theatre Game:* Point to different areas of an imaginary stage when the teacher calls them out.

EVALUATION (The teacher will select and adapt questions from Weeks 1-10, focusing on understanding and application. Examples:)

1. You want to create a tie-dye cloth with green circles. What two primary colours will you need to mix, and what tying technique will you use?
2. In a choir, if the Sopranos are singing the main tune, what are the Altos most likely singing?
3. During a play, an actor needs to pick up a letter from a table. Is that letter a costume or a prop?
4. Why should a microphone cable be taped to the floor backstage?
5. If you are on stage and the conductor makes a wide opening gesture with their arms, what are they likely telling you to do? (Sing louder/forte).

ASSIGNMENT

1. **(Study Preparation):** Organize your notes from this term. Read through them carefully to prepare for the end-of-term examination.
2. **(Practical/Portfolio):** Ensure your finished and dried tie-dye cloth is neatly pressed (ironed with help) and ready to be submitted for final assessment.

3. **(Written Reflection):** Write down which area you enjoyed most this term: **Fabric Decoration (Visual Art)** or **Choral Singing/Theatre (Performing Art)**. Give two reasons for your choice.
-



PRIMARY 6 BASIC DIGITAL LITERACY SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Advanced Robotics Concepts	Robotic sensors: ultrasonic, infrared, touch, light, temperature. Actuators: motors, servos. Robot programming logic. Obstacle avoidance. Line following basics. Robot competitions overview.
2	Physical Computing	Introduction to microcontrollers: Arduino basics. Input and output pins. LED control. Button inputs. Simple circuits. Safety with electronics. Real-world physical computing applications.
3	Machine Learning Basics	Understanding machine learning: computers learning from data. Supervised vs unsupervised learning. Training models with examples. Image recognition, text classification. Bias in AI. Practical ML demonstrations.
4	AI in Pictoblox	Pictoblox AI extensions. Image recognition projects. Speech recognition. Text-to-speech. Pose detection. Creating AI-powered games. Combining AI with robotics simulations.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Game Development Project I	Complete game development cycle. Game design document creation. Choosing game genre. Designing characters and levels. Planning game mechanics. Storyboarding. Asset creation or sourcing.
8	Game Development Project II	Implementing game in Scratch or Pictoblox. Programming game logic. Adding multiple levels. Score systems and lives. Game testing and debugging.

		Sound effects and music.
9	Technology and Society	Technology impact on daily life. Digital divide: access inequality. Technology careers overview. Ethical technology use. Environmental impact of technology. E-waste awareness. Sustainable technology practices.
10	Preparing for Digital Future (	Future technology trends: VR, AR, IoT, quantum computing. Skills for future: coding, creativity, critical thinking, collaboration. Continuous learning importance. Building digital portfolio. Resources for further learning. Transition to secondary school ICT.
11	REVISION	Comprehensive review of entire primary school digital literacy curriculum. Final capstone projects combining multiple skills. Presentations and demonstrations.
12	EXAMINATION	End of Third Term/Annual Examination and Primary School Final Examination (Theory and Practical)
13	CLOSING/GRADUATION	Report distribution, awards ceremony, graduation to secondary school

PRIMARY 6 DIGITAL LITERACY LESSON NOTES

THIRD TERM

WEEK 1: ADVANCED ROBOTICS CONCEPTS

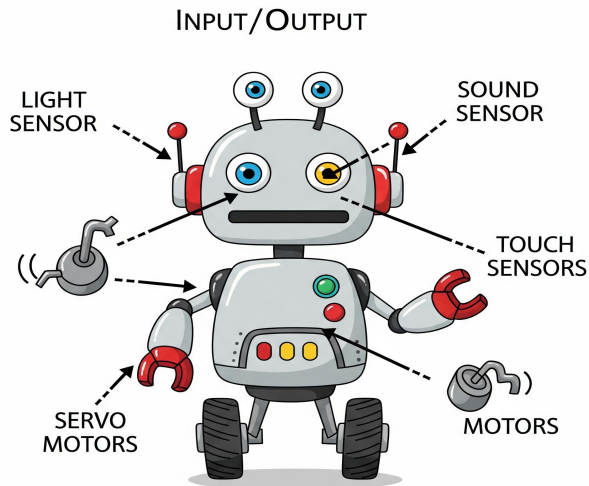
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define sensors and actuators in robotics.
2. List three types of robotic sensors and their functions.
3. Explain how a robot uses logic to avoid obstacles.
4. Describe the basic concept of a line-following robot.

CONTENT

1. Understanding Sensors and Actuators

- Think of a robot like a **human body**. To interact with the world, it needs two main things:
- **Sensors (The Robot's Senses - INPUT):** These are parts that **gather information** from the robot's surroundings, just like our eyes, ears, and skin.
 - *Example:* A sensor can "feel" if something is touching it, "see" how bright a room is, or "hear" a sound.
- **Actuators (The Robot's Muscles - OUTPUT):** These are parts that **create movement or action** based on commands from the robot's brain. They make things happen.
 - *Examples:*
 - **Motors:** Spin wheels to make the robot move forward, backward, or turn.
 - **Servos:** Allow for precise, controlled movement, like a robotic arm picking up a cup or a head turning side-to-side.



Sensors are the robot's senses.
Actuators are its muscles.

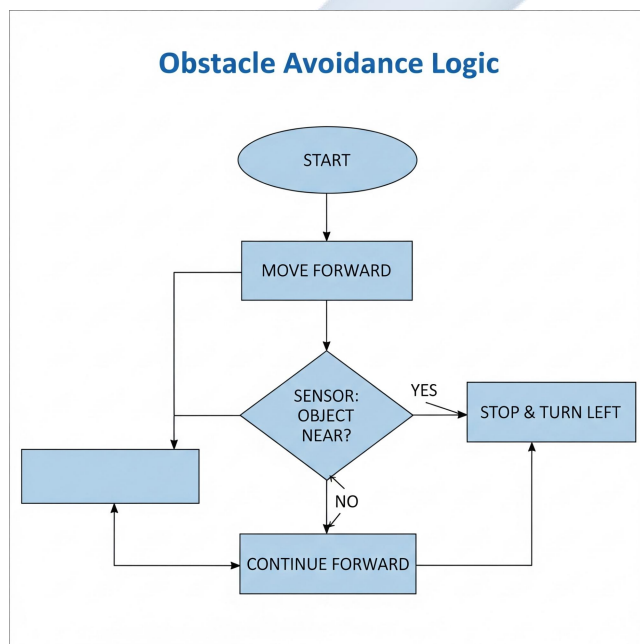
2. Types of Robotic Sensors

Robots use different sensors for different jobs:

- **Ultrasonic Sensor:** This works like a **bat**. It sends out high-pitched sound waves (too high for us to hear) and listens for the echo. If the echo comes back quickly, an object is **close**. If it takes longer, the object is **far**. It measures **distance**.
- **Infrared (IR) Sensor:** This uses invisible **light beams** to detect objects or colours. It is very common in TV remotes (to change channels) and in robots that follow lines on the floor.
- **Touch Sensor (Bumper Switch):** This is like a simple **button** or **whisker**. When the robot bumps into something, the button is pressed, and it sends a signal: "I've touched something!"
- **Light Sensor:** This measures the **brightness** of light in the room. It can tell the difference between a sunny day and a dark cupboard.
- **Temperature Sensor:** This can **feel heat and cold**, just like your skin. It can tell if an engine is getting too hot.

3. Robot Logic: Obstacle Avoidance

- Robots don't "think" for themselves. They follow a set of instructions called a **program** or **algorithm**. This is the robot's **logic**.
- Let's create the logic for a simple **Obstacle Avoidance Robot**:
 - a. **START**: The robot's program begins.
 - b. **ACTION**: The robot moves **FORWARD**.
 - c. **CHECK**: The ultrasonic sensor constantly checks: "**Is there an object in front of me closer than 20cm?**"
 - d. **DECISION**:
 - **IF YES** (Object is close) → **STOP** moving forward. **TURN LEFT** for 2 seconds. Then go back to Step 2.
 - **IF NO** (Path is clear) → **CONTINUE** moving forward. Go back to Step 3 to check again.



4. Line Following Basics

- A line-following robot is designed to follow a marked path, like a road.
- **How it is built**: It usually has **two infrared (IR) sensors** mounted side-by-side under its front, pointing at the ground.

- **The track:** A **black line** on a **white surface** (or vice versa).
- **The Logic:**
 - The IR sensor sends out light. Black absorbs light, white reflects it.
 - **If BOTH sensors see WHITE:** The robot is off the line. It must search.
 - **If LEFT sensor sees BLACK** (on the line) and **RIGHT** sees **WHITE:** The robot is drifting right. It corrects by turning **LEFT** a little.
 - **If RIGHT sensor sees BLACK** and **LEFT** sees **WHITE:** The robot is drifting left. It corrects by turning **RIGHT** a little.
 - This constant small correction keeps it wiggling along the line.

EVALUATION

1. A sensor is to a robot's brain as an _____ is to a robot's body. (Fill in the blank: sensor or actuator?)
2. Which part of a robot would make its wheels spin: a Sensor or an Actuator?
3. Which sensor uses sound waves like a bat to measure how far away something is?
4. What does an IR sensor use to detect objects or lines?
5. In the obstacle avoidance logic, what is the **FIRST** thing the robot should do after it senses a wall?
6. What colour combination is most common for a simple line-following robot track?
7. Give one real-life example of an actuator in a device at home (e.g., the spinning drum in a washing machine).

ASSIGNMENT

1. **(Drawing):** Draw your own simple robot design. Clearly **label** at least one place you would put a **sensor** and one place you would put an **actuator** (motor).
2. **(Observation/Written):** Look around your home. Find and write down the name of **one appliance** that you think uses a temperature sensor to work correctly (e.g., an electric iron, an air conditioner, an oven).
3. **(Logical Thinking/Written):** Complete this rule for a "Night Light Robot": "**IF** the light sensor detects that the room is dark, **THEN** the robot should turn on its _____."



WEEK 2: PHYSICAL COMPUTING

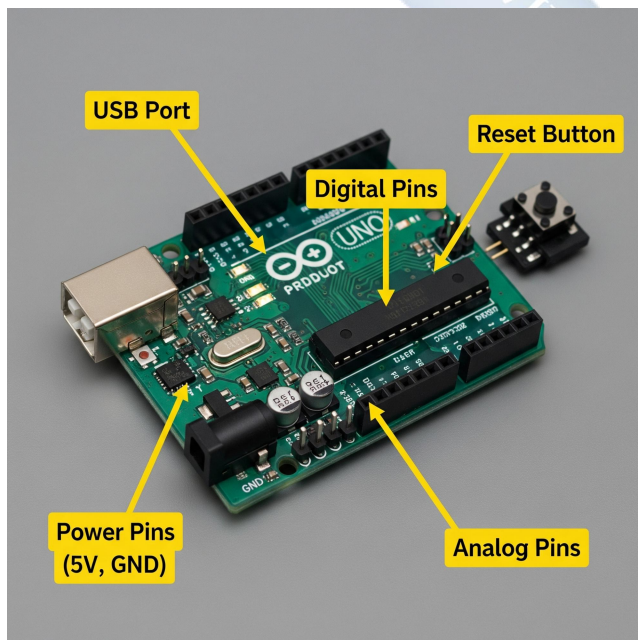
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define physical computing and microcontrollers.
2. Identify basic components like LEDs and Input/Output pins.
3. Explain how a simple circuit works with a microcontroller.
4. State important safety rules when working with electronics.

CONTENT

1. Introduction to Microcontrollers (Arduino)

- **Physical Computing** means making computers sense and control the **physical world** around us. Instead of just showing things on a screen, we make lights blink, motors spin, or sounds play based on real-world input.
- A **Microcontroller** is a tiny, simple computer on a single chip. It is not for browsing the internet, but it is perfect for reading sensors and controlling lights and motors. It's the "**brain**" of a physical computing project.
- **Arduino** is a very popular and easy-to-use microcontroller board for beginners. It has a USB port to connect to a computer for programming and power.

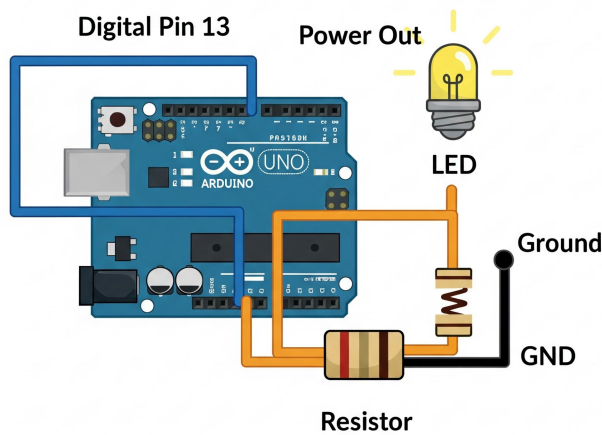


2. Inputs, Outputs, and Components

- **Pins:** These are the metal holes on the sides of the Arduino. They are like **ports** where we connect wires to other components.
 - **Digital Pins** (marked ~3, 5, 6, 9, 10, 11): Can be programmed as either INPUT (to read a button) or OUTPUT (to power an LED).
 - **Analog Pins** (marked A0-A5): Are mainly for INPUT from sensors that give a range of values (like a light sensor).
- **LED (Light Emitting Diode):** A small, efficient light. It has two legs:
 - **Longer leg (Anode):** Connects to **POSITIVE (+)** power.
 - **Shorter leg (Cathode):** Connects to **NEGATIVE (-) ground (GND)**.
- **Resistor:** A component that **limits the flow of electricity**. We use it with an LED to prevent too much current from flowing and burning out the LED.
- **Push Button:** A simple switch. When pressed, it completes a circuit, allowing electricity to flow.

3. Simple Circuits: Lighting an LED

- A circuit is a **complete loop** that electricity can flow around. If the loop is broken, the electricity stops.
- To light an LED with an Arduino:
 - a. Connect a wire from a **Digital Pin (e.g., Pin 13)** on the Arduino to the **long leg (+) of the LED**.
 - b. Connect a **Resistor** to the **short leg (-) of the LED**.
 - c. Connect another wire from the resistor to a **GND (Ground)** pin on the Arduino.
 - d. Write a simple program that sets Pin 13 to OUTPUT and turns it ON.
- **Electricity Path:** Power from Pin 13 → through wire → through LED → through resistor → back to GND → completes the loop.



4. Safety with Electronics (VERY IMPORTANT)

Working with electronics is fun but requires care:

1. **Dry Hands Only:** Never touch components or wires with wet hands. Water conducts electricity and can cause a short circuit or a mild shock.
2. **Disconnect Power:** Always **unplug the USB cable or remove the battery** before you change any wires or components. This prevents accidents.
3. **Handle Gently:** Do not force components into the board. The pins are delicate and can bend or break.
4. **Organised Workspace:** Keep your work area tidy. Loose wires can touch and cause a short circuit.

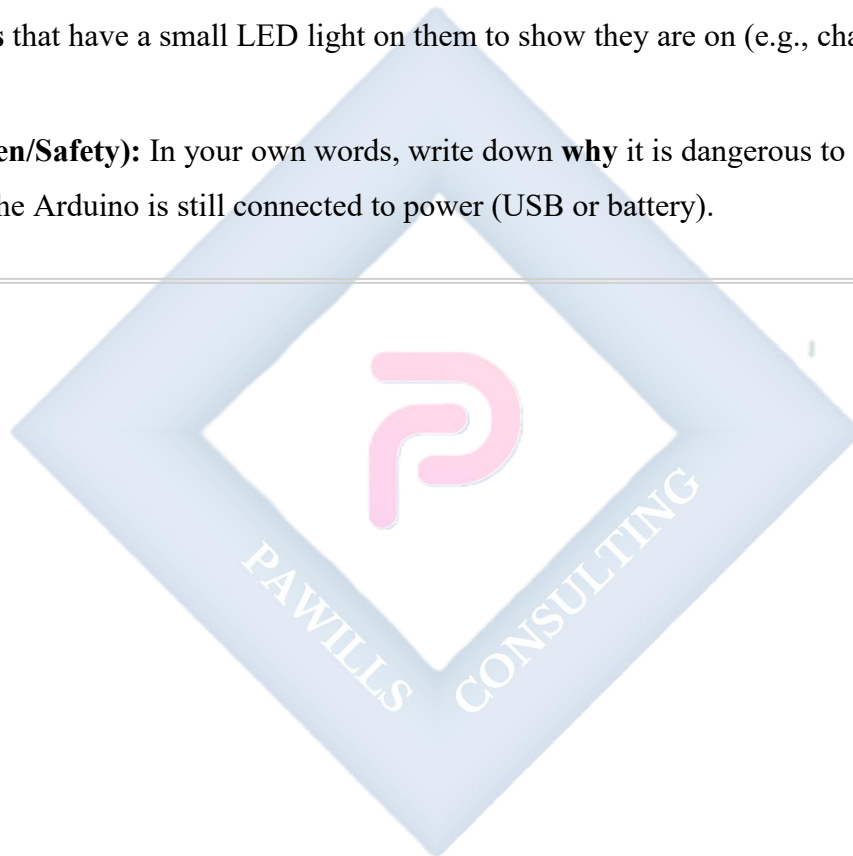
EVALUATION

1. What is the main job of a **microcontroller**?
2. Name the popular microcontroller board we use for learning physical computing.
3. If you want to read whether a button is pressed, should you connect it to an INPUT pin or an OUTPUT pin?
4. What does **LED** stand for? Which leg (long or short) connects to positive power?

5. What is the purpose of a **resistor** in an LED circuit?
6. What is the **MOST IMPORTANT** thing to do before you start changing wires on your Arduino project?

ASSIGNMENT

1. **(Drawing):** Draw the electronic symbol for an **LED**. (It looks like a triangle pointing to a line, with two arrows pointing away symbolizing light).
2. **(Observation/Oral):** Look around your house tonight. Find and list **two different devices** that have a small LED light on them to show they are on (e.g., charger, decoder, router).
3. **(Written/Safety):** In your own words, write down **why** it is dangerous to change wires while the Arduino is still connected to power (USB or battery).



WEEK 3: MACHINE LEARNING BASICS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Machine Learning in simple terms.
2. Differentiate between Supervised and Unsupervised learning.
3. Explain how computers recognize images using data.
4. Identify what "Bias" means in Artificial Intelligence (AI).

CONTENT

1. Understanding Machine Learning (ML)

- **Traditional Programming:** We give the computer **exact rules**. *"If the user presses 'A', THEN display 'Apple'."* The computer follows our instructions.
- **Machine Learning:** We give the computer **lots of examples and answers**, and it **figures out the rules for itself**. It learns from patterns in the data.
 - **Analogy:** How did you learn to recognize a dog? No one gave you a rulebook. You saw many pictures and were told "That's a dog." Your brain learned the pattern (four legs, fur, tail, etc.). Machine Learning works similarly for computers.

2. Types of Learning: With a Teacher vs. Without

- **Supervised Learning (Learning with a Teacher/Labels):**
 - We give the computer a **dataset where everything is already labeled**.
 - *Example:* We show it 1000 pictures, each with a tag: **"Cat"** or **"Dog"**.
 - The computer studies the pictures and learns the features of cats and dogs. Later, when we show it a **new picture**, it can guess the label based on what it learned.
- **Unsupervised Learning (Learning without a Teacher):**
 - We give the computer a **dataset with NO labels**.
 - *Example:* We give it a basket of 100 different fruits without names.
 - The computer's job is to **find patterns and group them** by itself. It might group all the round, red fruits together and all the long, yellow fruits together, but it won't know they are called "Apples" and "Bananas."

A SIMPLE TABLE COMPARING THE TWO TYPES

Supervised Learning	Unsupervised Learning
Teacher: Yes (Labeled Data)	Teacher: No (Unlabeled Data)
Goal: Learn to predict labels	Goal: Find hidden patterns/groups
Example: Email Spam Filter	Example: Customer Shopping Groups

3. How Computers "See" and "Read"

- **Image Recognition:** The computer doesn't see a picture like we do. It sees a grid of numbers representing colours. A Machine Learning model looks for patterns in these numbers.
 - *Process:* Show many "Cat" pictures → Model learns number patterns for "ears," "whiskers," "fur texture" → When a new picture comes, it checks for these patterns and says "This is 95% likely a cat."
- **Text Classification:** This is how your email knows what is spam.
 - *Process:* The model is shown thousands of emails marked "Spam" or "Not Spam." It learns that words like "**free**," "**winner**," "**click here**," "**urgent**" appear often in spam. When a new email arrives, it checks for these words and decides.

4. Bias in AI: A Critical Problem

- **What is Bias?** In AI, bias means the system makes **unfair or incorrect assumptions** because it was trained on limited or bad data.
- **Simple Example:** If you only train a face recognition system with pictures of light-skinned people, it might perform poorly or make mistakes when trying to recognize dark-skinned people. This is **bias**.
- **Why it matters:** AI should be fair and work well for **everyone**. To fix bias, we must train AI with **diverse, high-quality data** from all kinds of people and situations.

EVALUATION

1. What is the key difference between traditional programming and machine learning?
2. In **Supervised Learning**, what do we give the computer along with the examples?
3. Imagine you give a computer data on customer purchases with no categories. What type of learning (Supervised or Unsupervised) would it use to group similar customers together?
4. How does an email service use machine learning to filter spam?
5. What does "bias" mean in the context of an AI program?
6. If a voice assistant is only trained with American accents, what problem might a Nigerian user with a strong Igbo or Yoruba accent face?

ASSIGNMENT

1. **(Reflection/Written):** Think about your skills. List **three things** you learned by watching and practicing (like riding a bicycle or frying an egg) rather than by memorising rules from a book.
 2. **(Drawing/Analysis):** Draw a cluster of mixed shapes (e.g., triangles, stars, circles). Show with circles and arrows how a computer using **unsupervised learning** might group them into **two or three different clusters**.
 3. **(Critical Thinking/Written):** In one sentence, explain why it is important for the people who create AI to collect data from many different types of people all over Nigeria.
-

WEEK 4: AI IN PICTOBLOX

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify Pictoblox as a coding platform with AI features.
2. Describe how to use the Image Recognition extension.
3. Explain how Face Detection works in a program.
4. List creative ways AI can be used in games.

CONTENT

1. Introduction to Pictoblox

- **Pictoblox** is a **block-based coding software** (like Scratch) that has special superpowers: built-in **Artificial Intelligence (AI)** and **physical computing** tools.
- You can drag and drop code blocks to make projects that use your **computer's webcam** to see, recognise objects, detect faces, and even understand poses.

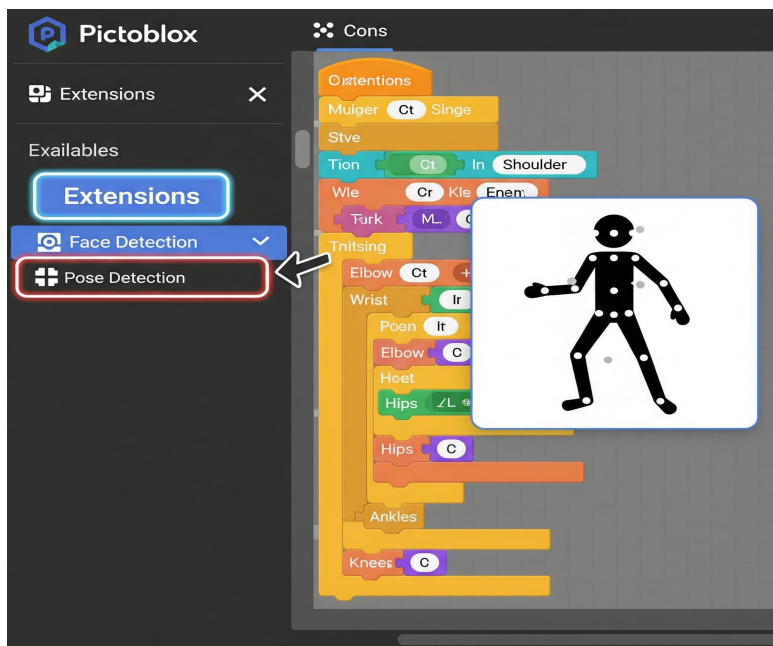
2. Image Recognition Projects

- **How to Start:** In Pictoblox, you add an **"Extension"** called **"Machine Learning"** or **"Computer Vision"**.
- **How it Works:** Once the extension is activated, you can use blocks like:
 - When video on [] is turned on (to start the camera).
 - If [camera] detects [object] (like a book, cup, or person).
- **Project Idea – Smart Desk Helper:** Code a program that uses the camera. **IF** it sees a **"book"** on your desk, **THEN** the computer character says, **"Time to read!"**. This uses a **pre-trained model** that already knows what common objects look like.

3. Face and Pose Detection

- **Face Detection:** The AI can find human faces in the camera view. It can even identify key points like eyes, nose, and mouth. More advanced models can guess the **emotion** (happy, sad, surprised).

- *Project Idea:* Make a "Mood Mirror." If it detects a **happy** face, it plays cheerful music. If it detects a **sad** face, it tells a joke.
- **Pose Detection:** This is amazing! The AI can track the position of your **body joints** (shoulders, elbows, wrists, hips, knees, ankles) in real-time.
 - *Project Idea – Virtual Controller:* Create a game where you control a character on screen by **raising your right hand to jump** or **leaning left/right to move**. No keyboard or mouse needed!



4. Speech and Text Features

- **Speech Recognition (Speech-to-Text):** You can speak into your microphone, and Pictoblox will convert your words into text on the screen. Block: When I hear [] → You say "Hello," and the sprite says "Hello back!"
- **Text-to-Speech:** You can make the computer **speak** any text you type. Block: Speak [Hello World]. It uses a robotic voice to read it aloud. Great for making talking assistants or storytellers.

EVALUATION

1. Pictoblox is similar to Scratch but has special tools for what new technology?

2. What do we call the add-on tools in Pictoblox that give it AI abilities?
3. How does **face detection** differ from **pose detection**? (Hint: What does each one track?)
4. Which AI feature would you use to make the computer type out the words you are saying?
5. Describe a simple game you could create where the player uses their **body pose** (not a mouse) to control a character.
6. True or False: To use image recognition in Pictoblox, you need to train the AI model yourself from scratch.

ASSIGNMENT

1. **(Planning/Written):** Write a simple **pseudocode script** for a program. "When the program starts, turn on the camera. If the camera detects a smiling face, make the sprite say 'You look happy!'."
 2. **(Observation/Written):** Look at your own face in a mirror. List **two** different emotions you can show (e.g., happy, sad, surprised, angry) that an advanced AI might try to detect.
 3. **(Analysis/Written):** In one sentence, explain why controlling a game with your **body** (pose detection) could be more fun or engaging than just pressing arrow keys on a keyboard.
-

WEEK 5 & 6: MIDTERM EXAMINATION AND BREAK



WEEK 7: GAME DEVELOPMENT PROJECT I (PLANNING)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the importance of planning before coding a game.
2. Create a simple Game Design Document (GDD).
3. Choose a genre and design characters for a game.
4. Draw a storyboard for a game level.

CONTENT

1. The Game Development Cycle

- Making a good game is like building a house. You **don't start by laying bricks**; you start with a **plan**. Professionals follow a cycle:
 - a. **PLAN (Design):** This is where we are now. We brainstorm, write ideas, and draw.
 - b. **BUILD (Code):** Next, we use software (Scratch/Pictoblox) to create the game.
 - c. **TEST (Play):** We play the game to find mistakes and see if it's fun.
 - d. **FIX (Debug):** We go back and improve the code and design based on testing.

2. Game Design Document (GDD) – Your Game's Blueprint

- A **GDD** is a simple document that answers the most important questions about your game. It keeps you focused.
 - **Title:** What is the cool name of your game? (e.g., "Lagos Rush", "Jollof Jump")
 - **Goal/Objective:** What must the player do to win? (e.g., "Collect 50 coins," "Rescue the stolen artefact," "Reach the finish line first.")
 - **Genre:** What type of game is it? Common types are:
 - **Platformer:** Jumping between platforms (like Super Mario).
 - **Puzzle:** Solving brain-teasers.
 - **Racing:** Driving to the finish line.
 - **Endless Runner:** Running forever while avoiding obstacles.

3. Characters and Mechanics

- **Characters:**

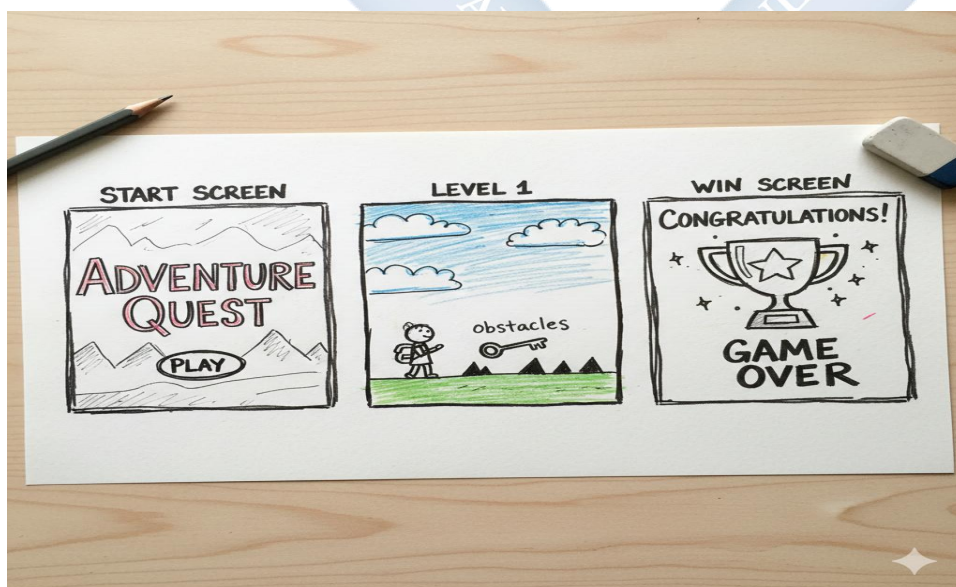
- **Protagonist:** The **hero** or main character the player controls. Give them a name and a reason for their quest.
- **Antagonist:** The **villain** or obstacle (e.g., a monster, a thief, a ticking clock).

- **Game Mechanics:** These are the **rules** and **actions** in the game.

- *Control Mechanics:* "Press SPACE to jump." "Use ARROW KEYS to move."
- *Interaction Mechanics:* "Collect coins to increase score." "Touch a red enemy to lose a life." "Open a chest to get a key."

4. Storyboarding – Drawing Your Game

- A **storyboard** is a series of simple drawings, like a comic strip, that shows what happens in the game from start to finish.
- It helps you plan the **visual flow**. Draw at least three key scenes:
 - a. **Title Screen:** The first screen with the game's name and a "Start" button.
 - b. **Gameplay Screen:** What the main level looks like. Show the hero, some coins, and an enemy.
 - c. **End Screen:** What happens when you win ("You Win!") or lose ("Game Over").



EVALUATION

1. What is the **very first step** in the game development cycle, and why is it so important?
2. What does the abbreviation **GDD** stand for?
3. How does writing a GDD help a game developer?
4. What are **game mechanics**? Give one example.
5. What is the purpose of creating a **storyboard** before coding?
6. Name two different **genres** of video games.

ASSIGNMENT

1. **(Written):** Create a **one-page Game Design Document** for your own game idea. It must include: **Game Title, Player's Goal, Genre, and Names of the Hero and Villain.**
2. **(Drawing):** On a sheet of paper, **draw a simple storyboard** with three panels showing the **Start, Middle (action), and End** of your game.
3. **(Critical Thinking/Written):** List **three specific rules (mechanics)** for your game. (e.g.,
1. Press UP ARROW to jump. 2. Collect 10 stars to unlock the door. 3. Avoid touching the spinning spikes.)

WEEK 8: GAME DEVELOPMENT PROJECT II (IMPLEMENTATION)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Translate a game plan into code using Scratch or Pictoblox.
2. Implement scoring and life systems in a game.
3. Add sound effects and background music.
4. Debug common errors in game logic.

CONTENT

1. Implementing Game Logic from Plan to Code

- Now, we take our GDD and storyboard and **build the game** in Scratch or Pictoblox.
- **Core Code Blocks:**
 - **forever loop:** This block keeps the main game actions running continuously. It's the game's heartbeat.
 - **if...then conditions:** These are used for **everything**. "If the player presses right arrow, **then** move right." "If the hero touches an enemy, **then** lose a life."

2. Creating Scores and Lives Systems

- **Variables** are essential for tracking game state.
 - **Creating a Score:**
 - i. Make a variable called Score.
 - ii. Set Score to 0 at the start of the game.
 - iii. When the player collects a coin, use the block: change [Score] by (1).
 - **Creating Lives:**
 - i. Make a variable called Lives.
 - ii. Set Lives to 3 at the start.
 - iii. When the hero touches an enemy: change [Lives] by (-1).
 - iv. **Game Over Check:** Always check if [Lives] = 0 then broadcast [Game Over] and switch to a "Game Over" screen.

3. Sound and Atmosphere – Making it Fun!

- Sound turns a moving picture into an engaging experience.
 - **Background Music:** A looping music track that plays throughout the game. Use the play sound [] until done block inside a forever loop or start it at the beginning.
 - **Sound Effects:** Short, sharp sounds for specific actions.
 - **Coin Collect:** A "ping" or "cha-ching" sound.
 - **Jump:** A "boing" sound.
 - **Game Over:** A sad trombone or "wah-wah" sound.
 - **Tip:** Balance the volume so the music doesn't drown out the effects.

4. Testing and Debugging – Be a Game Tester!

- **Playtesting:** The **most important** part of making a game. Play your own game over and over.
- **Common Bugs to Look For:**
 - **Character gets stuck** in walls (check your movement and collision code).
 - **Score doesn't reset** when you play again (make sure to set Score to 0 when the green flag is clicked).
 - **Sound plays too many times** (check if sound blocks are inside loops that run too fast).
- **Ask a Friend:** Have a classmate play your game. They will find problems you missed because they see it with fresh eyes. This is called **peer testing**.

EVALUATION

1. Which coding block makes a set of actions repeat forever, keeping the game running?
2. What type of code element (variable, loop, condition) would you use to keep track of the player's points?
3. Write a simple **if-then** rule for what should happen when the player character touches a "poison" object (lose a life).

4. What is the difference between **background music** and a **sound effect**? When would you use each?
5. Why is it very helpful to ask a friend to test the game you created?
6. If your game character is supposed to jump but doesn't, what is this problem called, and what should you do?

ASSIGNMENT

1. **(Pseudocode/Written):** Write the step-by-step logic (not specific blocks, just steps) for collecting a "power-up" in a game. (e.g., 1. Character touches power-up. 2. Power-up disappears. 3. Score increases by 10. 4. A "power-up" sound plays.)
2. **(Creative Thinking/Written):** Imagine you are making a **car racing game** set in Lagos. List **two different sound effects** you would need to record or find for your game.
3. **(Problem-Solving/Written):** Your game has a bug: when the player wins, the score doesn't reset to zero for the next round. What is one thing you would check in your code to fix this?

WEEK 9: TECHNOLOGY AND SOCIETY

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the concept of the "Digital Divide."
2. List three careers in technology.
3. Define E-waste and its impact on the environment.
4. Suggest ways to use technology sustainably.

CONTENT

1. Impact on Daily Life and The Digital Divide

- **Technology in Our Lives:** It helps us learn (online classes), communicate (WhatsApp, Zoom), bank (mobile apps), travel (ride-hailing), and even see a doctor (telemedicine).
- **The Digital Divide:** This is the **gap** between people who have easy access to computers and the internet and those who do not.
 - It can be between **cities and villages**, or between **wealthy and poor families**.
 - **Why it's a problem:** A child without internet at home cannot do online research or access digital lessons as easily as a child who can. This can create an unfair disadvantage in education and future jobs.
 - **The Goal:** To **bridge the divide** by providing affordable access, digital literacy training, and infrastructure (like internet towers) everywhere.

2. Exciting Careers in Technology

- The tech industry is full of amazing jobs for people who love solving problems and being creative:
 - **Software Developer/Engineer:** Writes the code for apps, websites, and systems we use every day.
 - **Robotics Engineer:** Designs, builds, and programmes robots for factories, hospitals, or exploration.
 - **Data Scientist:** Analyzes huge amounts of data (like market trends or health records) to find patterns and help make smart decisions.

- **Cybersecurity Analyst:** Protects computer systems and networks from hackers and cyber attacks.
- **Graphic/Game Designer:** Creates the visual art, animations, and user interfaces for games and apps.

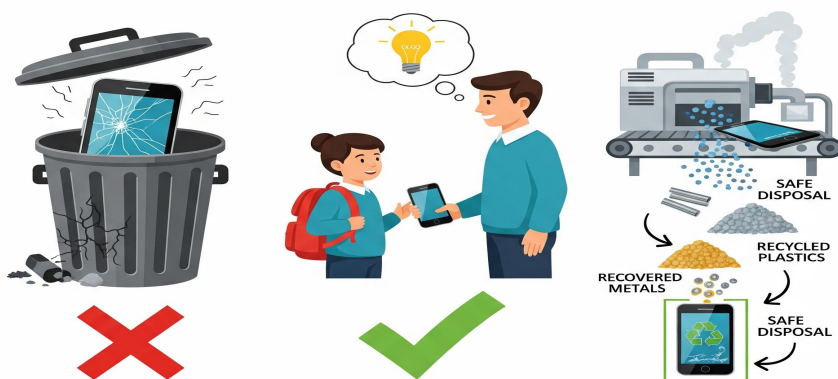
3. E-Waste (Electronic Waste) – A Growing Problem

- **What is E-waste?** Any discarded electronic device: old mobile phones, broken laptops, dead batteries, used printers, and old TV sets.
- **Why is it dangerous?** These devices contain **toxic materials** like lead, mercury, and cadmium. When thrown into regular dumpsites, these chemicals can **leak into the soil and water**, poisoning the environment and harming people's health.
- **The Scale:** Nigeria generates hundreds of thousands of tonnes of e-waste every year, often from imported used electronics that finally break down.

4. Sustainable Technology Practices – What Can We Do? We must follow the **3 Rs** for electronics:

- **Reduce:** Think before you buy. Do you really need the newest phone if your old one works fine? Using devices longer reduces waste.
- **Reuse:** If a device still works but you don't need it, **donate it**. Give an old tablet or computer to a younger sibling, a neighbour, or a community learning centre.
- **Recycle:** If it is completely broken, **do not put it in the normal bin**. Find a proper **e-waste recycling centre**. They can safely take it apart and recover valuable materials like gold, copper, and plastic.

The Life of an Old Phone



**Donate or Recycle.
Never Trash E-waste**

EVALUATION

1. What is the "**Digital Divide**"? Give one example of how it can affect a pupil's education.
2. Name two technology careers that involve **writing code**.
3. What does the "**E**" in E-waste stand for?
4. Why is it harmful to throw an old phone battery into a regular rubbish dump?
5. What is one better option than throwing away a working but old laptop?
6. How has technology made it easier for a farmer in a rural area to sell his crops?

ASSIGNMENT

1. **(Research/Oral):** Ask a parent or guardian: "**What do we usually do with our old or broken phones in this house?**" Write down their answer.
2. **(Drawing/Creative):** Design a **poster** to encourage pupils in your school to recycle e-waste. Include a catchy slogan like "Don't Trash Tech, Recycle It!".
3. **(Reflection/Written):** Write **two sentences** about what you would like to be when you grow up. Does it involve technology? How?

WEEK 10: PREPARING FOR THE DIGITAL FUTURE

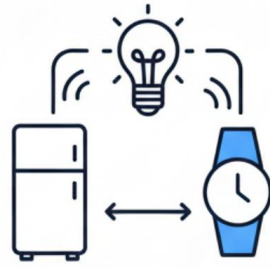
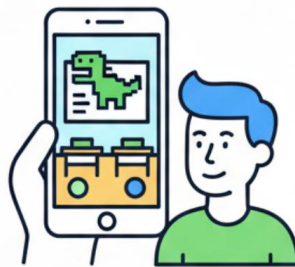
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify future technology trends (VR, AR, IoT).
2. List key skills needed for the future workplace.
3. Explain the importance of a Digital Portfolio.
4. Discuss the transition to Secondary School ICT.

CONTENT

1. Future Technology Trends

- **VR (Virtual Reality):** Using a special headset to be **fully immersed** in a computer-generated world. You can look around and feel like you're inside a video game, a museum tour, or a training simulation.
- **AR (Augmented Reality):** **Overlaying** digital images or information onto the real world you see. Examples: The **Pokémon GO** game where creatures appear in your street, or furniture apps that let you see how a new sofa would look in your living room.
- **IoT (Internet of Things):** Everyday objects connected to the internet, able to send and receive data. They become "smart."
 - *Examples:* A **smart fridge** that can tell you when the milk is finishing. **Smart streetlights** that brighten when they sense a car. **Wearable fitness trackers** that monitor your heart rate.



2. Skills for the Future Workplace

Technology will change jobs, but these **human skills** will always be in demand:

- **Creativity:** The ability to **invent new ideas**, solutions, and content. Robots follow rules; humans imagine new things.
- **Critical Thinking:** The ability to **analyse information, solve complex problems, and make good decisions**. Asking "why" and "how."
- **Collaboration:** Working **effectively with others** in a team, both in person and online across distances.
- **Communication:** Clearly **explaining your ideas** in writing and speaking.
- **Continuous Learning:** The willingness to **keep learning new skills** throughout your life because technology evolves so fast.

3. Building a Digital Portfolio

- A **Digital Portfolio** is a **curated collection of your best digital work**. It's like a showcase or CV that proves what you can do.
- **What to Include:**
 - Screenshots or links to your best **Scratch/Pictoblox games and animations**.
 - Well-formatted **Word documents or PowerPoint presentations** from projects.
 - Photos or videos of any **robotics or science projects**.
 - Certificates from coding competitions or online courses.
- **Where to Keep it:** Save it in an organised folder on **Google Drive** or **Microsoft OneDrive**. This way, you can access it from anywhere and easily share it with future teachers.

4. Transition to Secondary School ICT

- Congratulations! You are finishing Primary School with a strong foundation in Digital Literacy.
- In **Secondary School**, your ICT classes will build on what you have learned:
 - You will dive deeper into **text-based programming** (like Python).

- You will learn more about **computer hardware, networking, and data management**.
- You will use more advanced software for **graphics, web design, and presentations**.
- **Your Primary 6 knowledge is the key that opens the door to these more advanced topics. Be confident!**

EVALUATION

1. If you put on a headset and feel like you are walking on the moon, are you experiencing **VR** or **AR**?
2. Explain the **Internet of Things (IoT)** with an example of a device you might find in a "Smart Home."
3. Which future skill involves working well with other people to achieve a common goal?
4. What is a **Digital Portfolio** and why is it useful for a student?
5. Why is "Continuous Learning" especially important in the field of technology?
6. What is one new ICT topic you are excited to learn about in Secondary School?

ASSIGNMENT

1. **(Practical/Written):** Look back at your work from this term and last term. List **three specific items** you would proudly include in your Digital Portfolio (e.g., "My Python password checker code", "My game design document for 'Lagos Rush'").
 2. **(Imagination/Written):** Imagine you live in a "Smart Home" in 2030. Describe **one smart object** in your bedroom and what it does automatically to help you (e.g., "A smart window that closes automatically when it senses rain.").
 3. **(Reflection/Written):** Write a **short note** (3-4 sentences) to your future Secondary School ICT teacher. Introduce yourself and mention **two digital literacy skills** you learned in Primary 6 that you think will be helpful.
-

WEEK 11: REVISION

CONTENT

1. Comprehensive review of Third Term topics:

- **Robotics:** Sensors (Ultrasonic, IR, Touch) vs. Actuators (Motors, Servos). Simple robot logic (Obstacle Avoidance, Line Following).
- **Physical Computing:** Microcontrollers (Arduino), basic components (LEDs, resistors), simple circuits, and **safety rules**.
- **AI & Machine Learning:** Difference from normal programming. Supervised vs. Unsupervised Learning. Understanding **Bias** in AI. Using AI in **Pictoblox** (Face/Pose Detection, Speech).
- **Game Development:** The planning cycle (GDD, Storyboarding) and implementation (Variables for score/lives, adding sound, debugging).
- **Technology & Society:** The Digital Divide, Tech Careers, **E-waste problems and solutions** (Reduce, Reuse, Recycle).
- **The Digital Future:** Trends (VR, AR, IoT), essential future skills, and the importance of a **Digital Portfolio**.

2. Final Presentation of Projects:

- Pupils will have time to put final touches on their **Game Development Projects** from Weeks 7 & 8.
- Each pupil or group will have 3-5 minutes to **demonstrate their game** to the class, explaining its goal, mechanics, and what they learned during creation.

3. Demonstration of Key Concepts:

- Teacher-led recap with physical components: Show an Arduino, LED, and sensor.
- Live demonstration of a simple Pictoblox project using face or pose detection.

4. General Q&A and Exam Preparation:

- Open floor for pupils to ask any remaining questions about all topics from the Third Term or the entire year.
- Guidance on how to prepare for end-of-term and common entrance examinations, focusing on understanding concepts rather than rote memorisation.